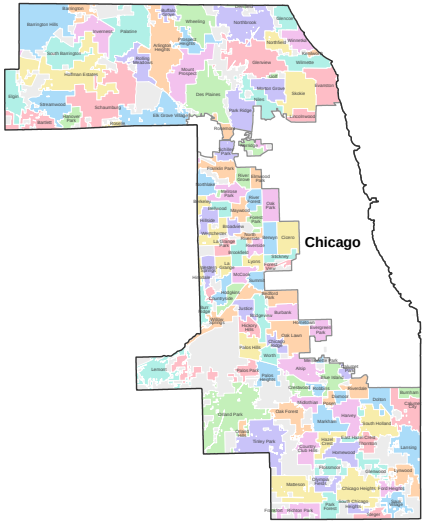
Breaking Away:
The Endogenous Formation of Local Governments

Jie Song

University of California, Berkeley

Nov 6, 2026

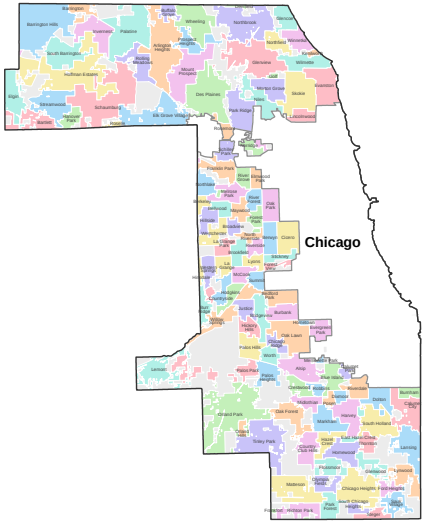
Why are local governments formed?



134 municipalities

in Cook County, Illinois

Why are local governments formed?

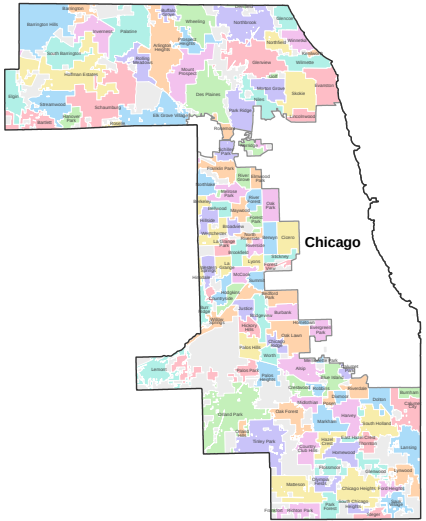


134 municipalities

in Cook County, Illinois

Municipalities: set own municipal public goods, taxes and zoning

Why are local governments formed?



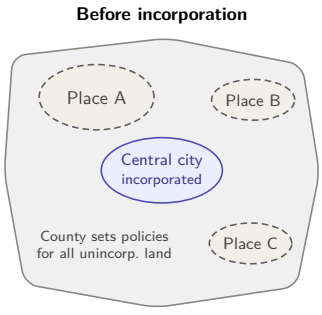
134 municipalities

in Cook County, Illinois

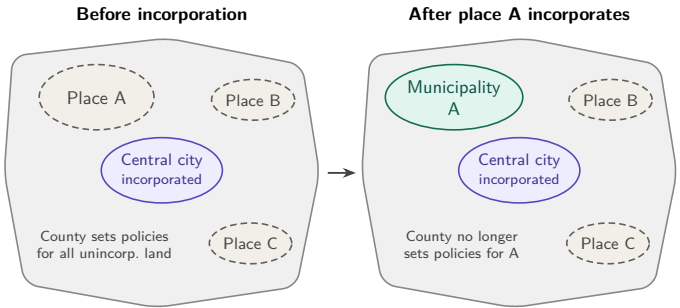
Municipalities: set own municipal public goods, taxes and zoning

County: directly governs the unincorporated land (hatched)

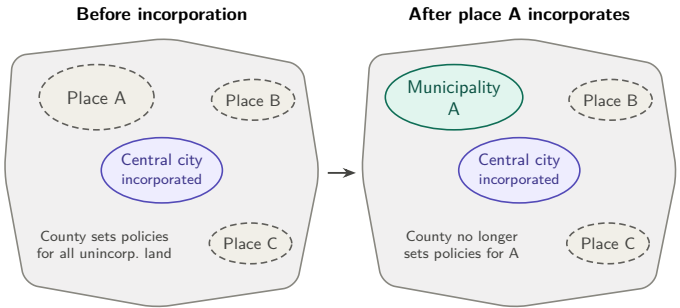
Municipal incorporation changes who sets policies



Municipal incorporation changes who sets policies



Municipal incorporation changes who sets policies

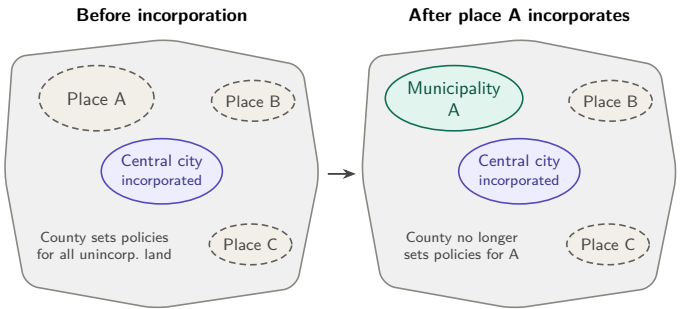


The process

- **Petition**: 10% (Washington) to 25% (California) of voters
- **Institutional review**: relevant agency approval
- **Referendum**: voters decide
- **Legislature approval** required in some states

Difficulty varies by state → the formation cost of a new government

Municipal incorporation changes who sets policies



The process

- **Petition**: 10% (Washington) to 25% (California) of voters
- **Institutional review**: relevant agency approval
- **Referendum**: voters decide
- **Legislature approval** required in some states

Difficulty varies by state → the formation cost of a new government

Gains from policies + tax base > formation cost ⇒ incorporation

This paper

Quantifies what residents gain when they form their own government

This paper

Quantifies what residents gain when they form their own government

1. **Empirical context and motivating facts**
   - 353 incorporations in U.S. metropolitan areas, 1970–2020; histories of 880 attempts, with referendum votes and stated motives
   - A comprehensive dataset: residential composition, and local policies of places and the county
   - Four motivating facts: who forms their own governments, and what changes afterward?

This paper

Quantifies what residents gain when they form their own government

1. **Empirical context and motivating facts**
   - 353 incorporations in U.S. metropolitan areas, 1970–2020; histories of 880 attempts, with referendum votes and stated motives
   - A comprehensive dataset: residential composition, and local policies of places and the county
   - Four motivating facts: who forms their own governments, and what changes afterward?

2. **Quantitative model**
   - How much do residents gain and from which channel?
   - Distribution of gains by resident type, inside and outside the incorporator
   - Counterfactuals: who must consent, and whether the new city can compensate the remainder

Preview of results

Four motivating facts	Model and estimation

Preview of results

	Four motivating facts	Model and estimation
1	Local policies vary with constituency	$\Rightarrow$ Heterogeneity in policy valuation $\Rightarrow$ Discipline preference parameters

Preview of results

	Four motivating facts	Model and estimation
1	Local policies vary with constituency	$\Rightarrow$ Heterogeneity in policy valuation $\Rightarrow$ Discipline preference parameters
2	Whiter, richer, and larger places incorporate	$\Rightarrow$ Discipline formation parameters

Preview of results

	Four motivating facts	Model and estimation
1	Local policies vary with constituency	⇒ Heterogeneity in policy valuation ⇒ Discipline preference parameters
2	Whiter, richer, and larger places incorporate	⇒ Discipline formation parameters
3	Spending mix and new construction diverge between incorporator and county post incorporation	⇒ Source of policy-fit gains

Preview of results

	Four motivating facts	Model and estimation
1	Local policies vary with constituency	⇒ Heterogeneity in policy valuation ⇒ Discipline preference parameters
2	Whiter, richer, and larger places incorporate	⇒ Discipline formation parameters
3	Spending mix and new construction diverge between incorporator and county post incorporation	⇒ Source of policy-fit gains
4	Population growth follows incorporation, college share increases	⇒ Benchmark for residential mobility

Preview of results (cont'd)

	What it means
Preferences	
Zoning valuation	Differs mainly by race: White residents prefer minimum lots of 22,600–26,900 sq ft, Other residents 1,700–6,900 sq ft
Public goods valuation	Similar across types: preferred property tax rate of 15–22% on housing payments (about 1–1.4% of home value a year)

Preview of results (cont'd)

	What it means
Preferences	
Zoning valuation	Differs mainly by race: White residents prefer minimum lots of 22,600–26,900 sq ft, Other residents 1,700–6,900 sq ft
Public goods valuation	Similar across types: preferred property tax rate of 15–22% on housing payments (about 1–1.4% of home value a year)
Formation	
Gains	A gain of 0.1% of income raises the odds of incorporating by 5.1%
Formation cost	Falls with size: doubling a place's households lowers it by 0.52% of income

Preview of results (cont'd)

	What it means
Preferences	
Zoning valuation	Differs mainly by race: White residents prefer minimum lots of 22,600–26,900 sq ft, Other residents 1,700–6,900 sq ft
Public goods valuation	Similar across types: preferred property tax rate of 15–22% on housing payments (about 1–1.4% of home value a year)
Formation	
Gains	A gain of 0.1% of income raises the odds of incorporating by 5.1%
Formation cost	Falls with size: doubling a place's households lowers it by 0.52% of income
Results	
Gain, average resident	New city **+0.22%** of income: tax base 0.10%, zoning net of housing cost 0.12%, public goods $<0.01\%$; remainder **−0.013%**

Preview of results (cont'd)

	What it means
Preferences	
Zoning valuation	Differs mainly by race: White residents prefer minimum lots of 22,600–26,900 sq ft, Other residents 1,700–6,900 sq ft
Public goods valuation	Similar across types: preferred property tax rate of 15–22% on housing payments (about 1–1.4% of home value a year)
Formation	
Gains	A gain of 0.1% of income raises the odds of incorporating by 5.1%
Formation cost	Falls with size: doubling a place's households lowers it by 0.52% of income
Results	
Gain, average resident	New city **+0.22%** of income: tax base 0.10%, zoning net of housing cost 0.12%, public goods $<0.01\%$; remainder **−0.013%**
Counterfactuals	Double consent stops *affluent* places (**−52%**) With compensation, most of them still incorporate (**−8%**)

Contribution

1. Jurisdiction formation.
Alesina and Spolaore (1997), Bolton and Roland (1997), Alesina et al. (2004), Gravel and Oddou (2014), Desmet et al. (2011), Weese (2015), Miller (1981), and Burns (1994)

$\Rightarrow$ **Quantify the motives** in a spatial model with endogenous local policies

Closest papers Key references by strand

Contribution

1. Jurisdiction formation.
Alesina and Spolaore (1997), Bolton and Roland (1997), Alesina et al. (2004), Gravel and Oddou (2014), Desmet et al. (2011), Weese (2015), Miller (1981), and Burns (1994)

⇒ **Quantify the motives** in a spatial model with endogenous local policies

2. Decentralization.
Besley and Coate (2003), Calabrese et al. (2012), Bordeu (2025), Ruggieri (2025), and Dahis and Szerman (2026)

⇒ **Endogenize decentralization decision,** given consequences of decentralization

Closest papers | Key references by strand

Contribution

1. Jurisdiction formation.
Alesina and Spolaore (1997), Bolton and Roland (1997), Alesina et al. (2004), Gravel and Oddou (2014), Desmet et al. (2011), Weese (2015), Miller (1981), and Burns (1994)

⇒ **Quantify the motives** in a spatial model with endogenous local policies

2. Decentralization.
Besley and Coate (2003), Calabrese et al. (2012), Bordeu (2025), Ruggieri (2025), and Dahis and Szerman (2026)

⇒ **Endogenize decentralization decision,** given consequences of decentralization

3. land use regulation.
Fischel (2001), Parkhomenko (2023), Bartik et al. (2025), Rollet (2025), Song (2025), and Gallagher et al. (2026)

⇒ **Land use motive in local government creation**: valued against other motives

Closest papers | Key references by strand

Roadmap

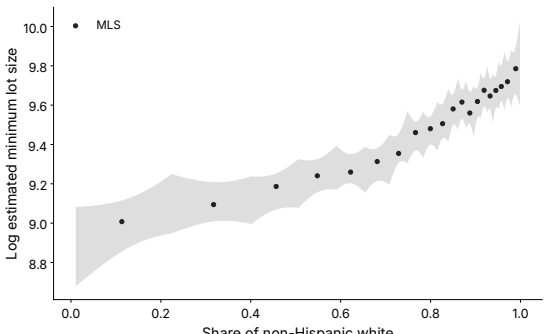
1. Motivating Facts
2. Model
3. Estimation
4. Gains from incorporation
5. Counterfactuals and policy implications

Data: U.S. municipal incorporation, 1970–2020

Economic object	**Key Measures**	**Sources**
Incorporation	353 incorporations; 880 attempts + results, votes and stated motives	Boundary and Annexation Survey incorporation histories
Constituency	Demographics in 2020 geography	Decennial census/ American Community Surveys, 1970–2020
Municipal public goods	Spending on public safety, infrastructure, and amenities	Census of Governments, 1977–2022
Zoning and land use	Minimum lot size, new single-family lots, house prices	County assessor records (ATTOM)

1 2 3 4 Local policies differ with constituency

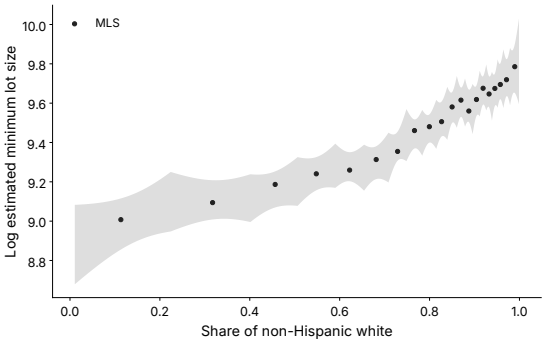
Minimum lot size ↑ in % non-Hispanic White and % college residents

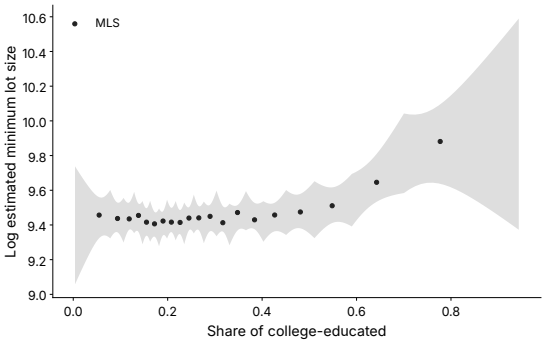


MLS measure | Table

❶ ② ③ ④ Local policies differ with constituency

Minimum lot size ↑ in % non-Hispanic White and % college residents

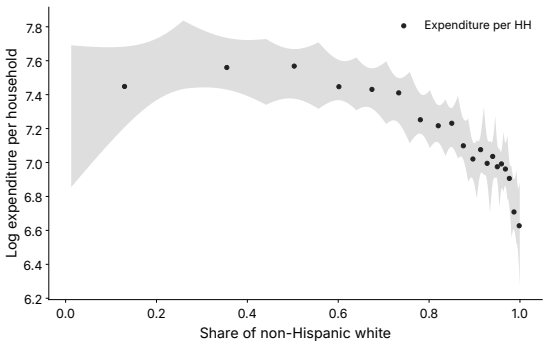




MLS measure | Table

1 2 3 4 Local policies differ with constituency

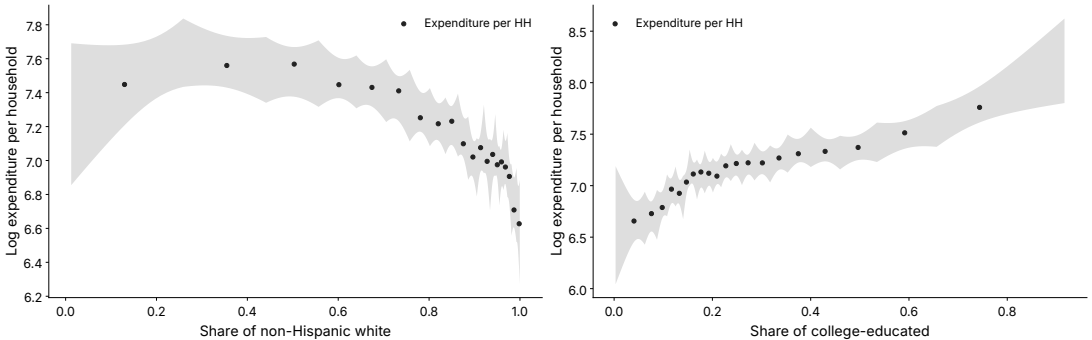
Spending per household ↓ in % non-Hispanic White, ↑ in % college residents



Tax base vs. tax rate

Local policies differ with constituency

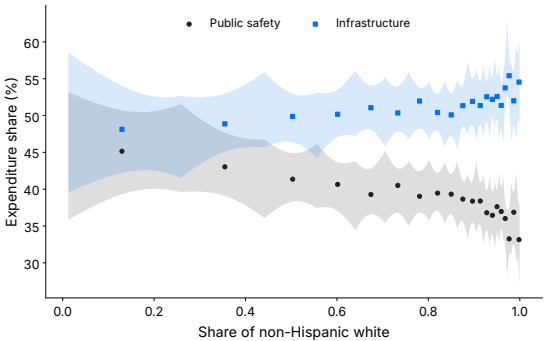
Spending per household ↓ in % non-Hispanic White, ↑ in % college residents



Tax base vs. tax rate

Local policies differ with constituency

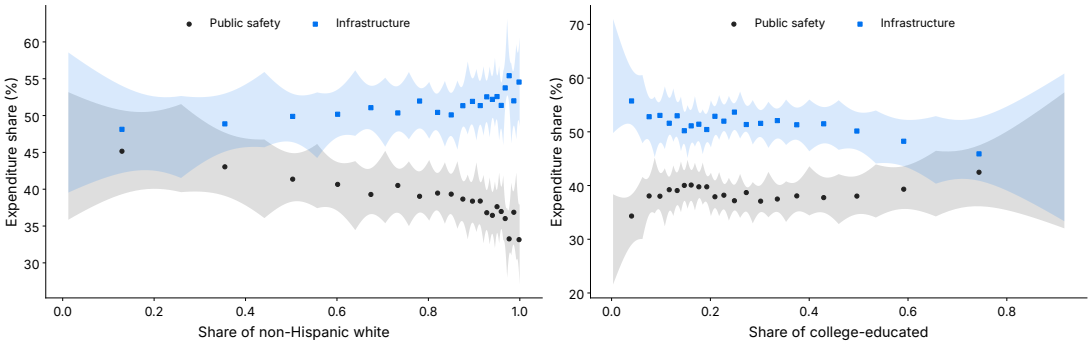
Municipal public good mixes also differ by constituency



Table

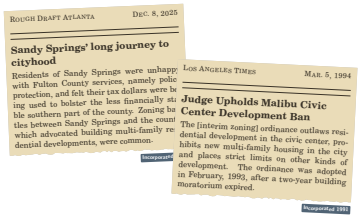
Local policies differ with constituency

Municipal public good mixes also differ by constituency



Table

Motives for incorporation

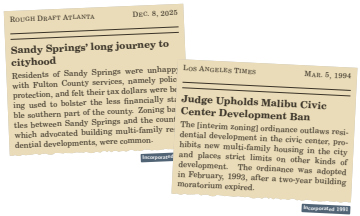


Sandy Springs, GA (Incorp. 2005)

- Better police than Fulton County provided

Stated motives

Motives for incorporation

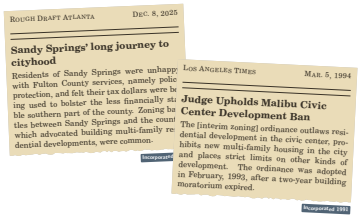


Sandy Springs, GA (Incorp. 2005)

- Better police than Fulton County provided
- Keep tax dollars from subsidizing south Fulton

Stated motives

Motives for incorporation

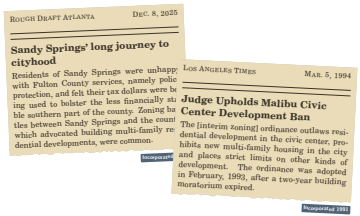


Sandy Springs, GA (Incorp. 2005)

- Better police than Fulton County provided
- Keep tax dollars from subsidizing south Fulton
- Stop county plans for apartments

Stated motives

Motives for incorporation



Sandy Springs, GA (Incorp. 2005)

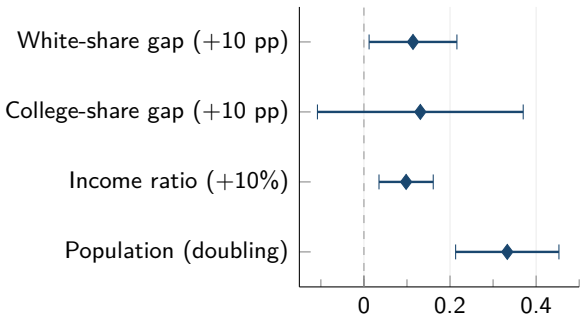
- Better police than Fulton County provided
- Keep tax dollars from subsidizing south Fulton
- Stop county plans for apartments

Malibu, CA (Incorp. 1991)

- Block the county sewer plan

Stated motives

Whiter, richer, and larger places incorporate

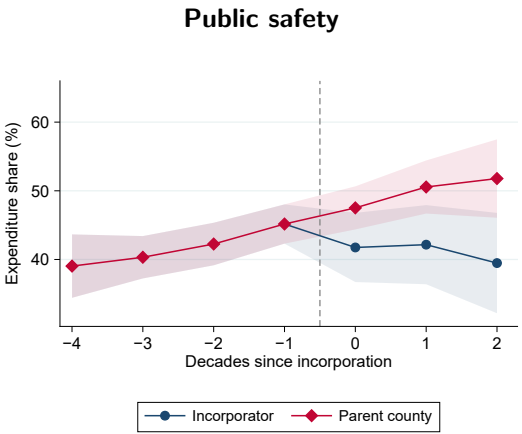


Linear probability model: predictors of incorporation

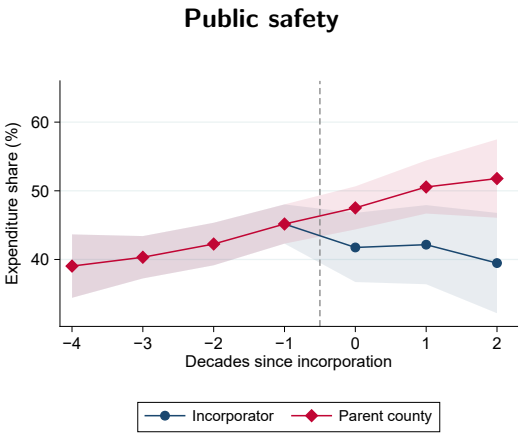
unincorporated place-decade panel;
gap = place minus its unincorporated county, lagged;
state-by-decade FE

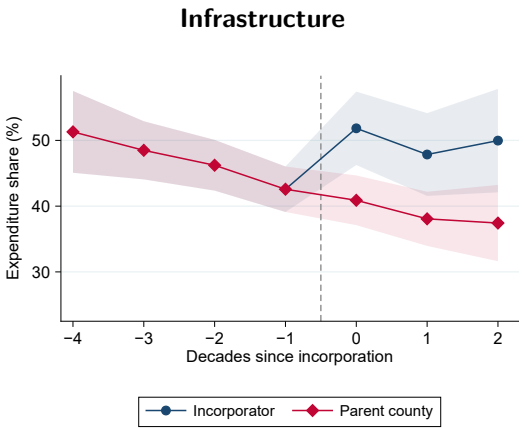
Table

After incorporation, spending mixes diverge



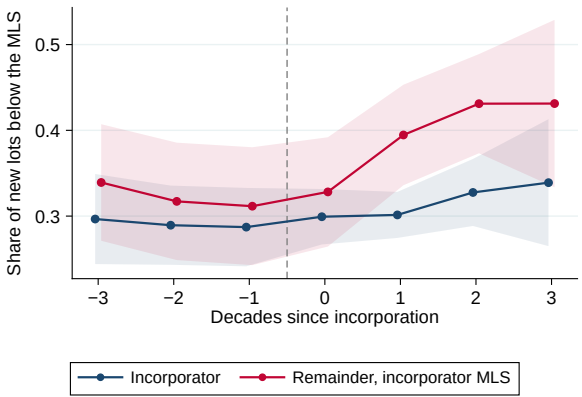
After incorporation, spending mixes diverge





New construction diverges: the remainder builds more small lots

Share of new single-family lots below the incorporator's minimum lot size

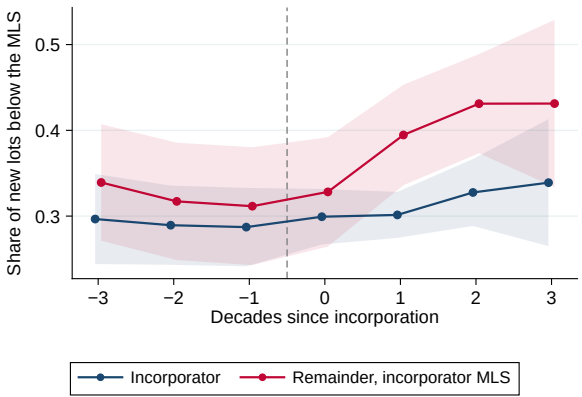


Incorporator

About 30% of new lots fall below its minimum lot size, before and after incorporation

New construction diverges: the remainder builds more small lots

Share of new single-family lots below the incorporator's minimum lot size



Incorporator

About 30% of new lots fall below its minimum lot size, before and after incorporation

Remainder

The share rises by about 10 pp a decade after the incorporator leaves

Consequences of incorporation: empirical design

Incorporators vs. failed attempts

- 326 incorporators in 38 states vs. 141 places whose formal attempt failed and that had not incorporated by 2020

Consequences of incorporation: empirical design

Incorporators vs. failed attempts

- 326 incorporators in 38 states vs. 141 places whose formal attempt failed and that had not incorporated by 2020
- Event study with an imputation estimator (Borusyak et al. 2024)

Consequences of incorporation: empirical design

Incorporators vs. failed attempts

- 326 incorporators in 38 states vs. 141 places whose formal attempt failed and that had not incorporated by 2020
- Event study with an imputation estimator (Borusyak et al. 2024)
- Counterfactual only estimated from later incorporators and failed attempts

Consequences of incorporation: empirical design

Incorporators vs. failed attempts

- 326 incorporators in 38 states vs. 141 places whose formal attempt failed and that had not incorporated by 2020
- Event study with an imputation estimator (Borusyak et al. 2024)
- Counterfactual only estimated from later incorporators and failed attempts
- Untreated potential outcomes: $Y_{it}(0) = \alpha_i + \upsilon_{state(i) \times t} + \varepsilon_{it}$

Consequences of incorporation: empirical design

Incorporators vs. failed attempts

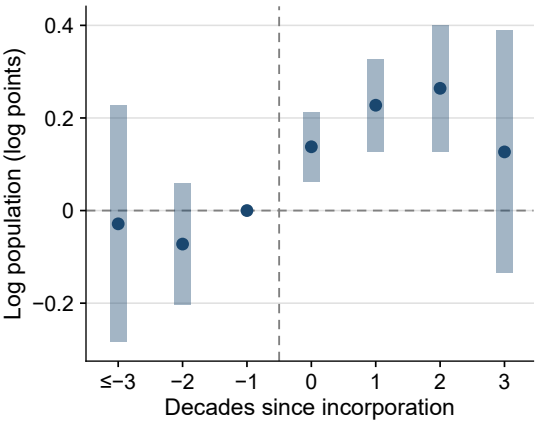
- 326 incorporators in 38 states vs. 141 places whose formal attempt failed and that had not incorporated by 2020
- Event study with an imputation estimator (Borusyak et al. 2024)
- Counterfactual only estimated from later incorporators and failed attempts
- Untreated potential outcomes: $Y_{it}(0) = \alpha_i + \upsilon_{state(i) \times t} + \varepsilon_{it}$
  - α_i: place FE, $\upsilon_{state(i) \times t}$: state $\times$ decade FE; standard errors clustered by state

Consequences of incorporation: empirical design

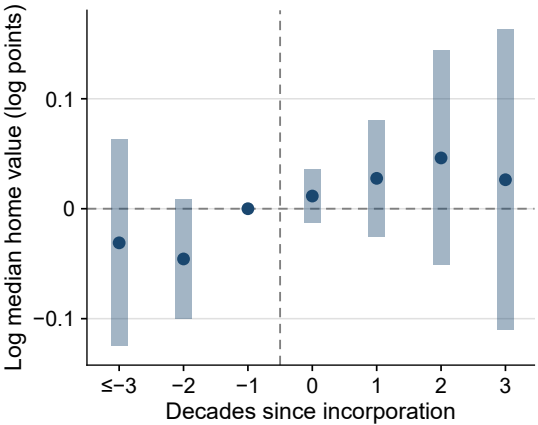
Incorporators vs. failed attempts

- 326 incorporators in 38 states vs. 141 places whose formal attempt failed and that had not incorporated by 2020
- Event study with an imputation estimator (Borusyak et al. 2024)
- Counterfactual only estimated from later incorporators and failed attempts
- Untreated potential outcomes: $Y_{it}(0) = \alpha_i + \upsilon_{state(i) \times t} + \varepsilon_{it}$
  - α_i: place FE, $\upsilon_{state(i) \times t}$: state $\times$ decade FE; standard errors clustered by state
  - Robustness: baseline population and gaps with the county, each interacted with decade

Incorporators see population growth post incorporation



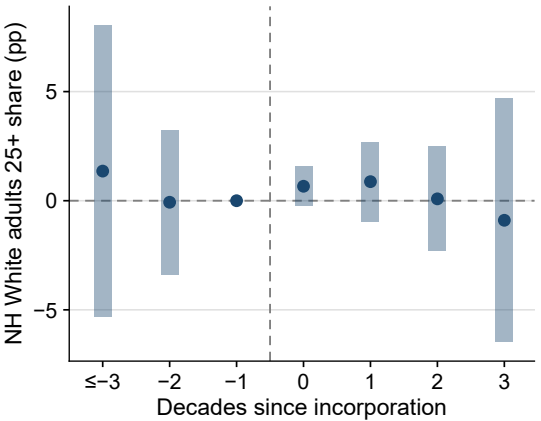
$\hat{\tau}$: $+0.21$ in logs (SE 0.05), about $+23\%$



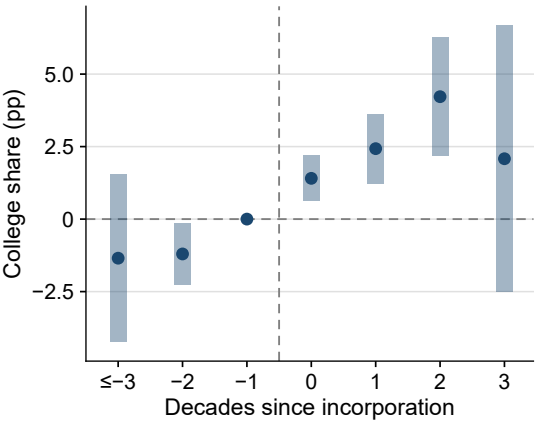
$\hat{\tau}$: $+0.03$ in logs (SE 0.03)

Table

College share increases by 2.7 pp



$\hat{\tau}$: +0.5 pp (SE 0.8)



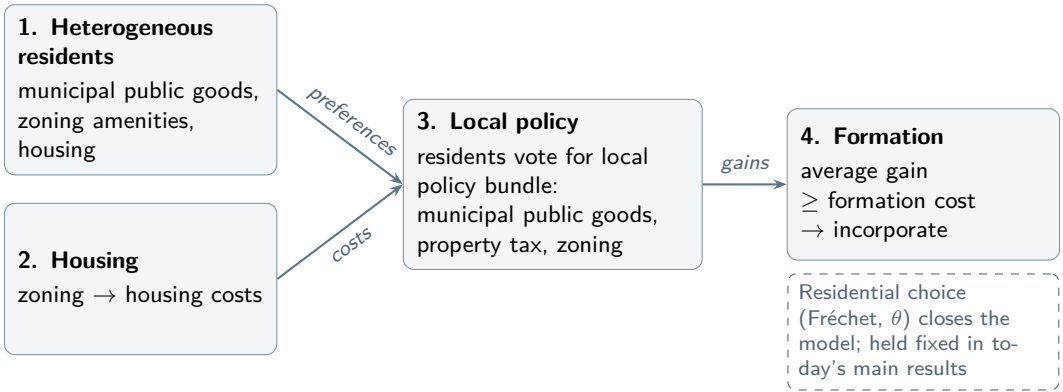
$\hat{\tau}$: +2.7 pp (SE 0.6)

Table

Roadmap

1. Motivating Facts
2. Model
3. Estimation
4. Gains from incorporation
5. Counterfactuals and policy implications

The model in four blocks



Residents differ in what they value

Utility of household ν of type s in location i of jurisdiction m

$$U_{i\nu} = \underbrace{c_\nu^{1-\beta_s}}_{\text{numeraire}} \underbrace{h_\nu^{\beta_s}}_{\text{housing}} \underbrace{\prod_k g_{mk}^{\gamma_{sk}}}_{\text{municipal public goods}} \underbrace{z_m^{\rho_s}}_{\text{zoning amenity}} \underbrace{B_{is}\,\epsilon_{i\nu}}_{\text{location tastes}}$$

$$c_\nu + \left(1 + \underbrace{t_m(\mathbf{g}_m)}_{\text{property tax}}\right) q_i(z_m)\, h_\nu = y_\nu$$

β_s: housing exp share

γ_{sk}: value of municipal public good k (public safety, infrastructure, amenities)

ρ_s: amenity value of zoning (open space, less congestion)

Zoning raises housing costs

- Supply: housing produced using fixed land and a variable input; zoning lowers productivity (e.g.,D'Amico et al. (2024))

$$H_i^S = \mathcal{A}_i e^{-(1+\eta)\kappa z_m} \left(q_i / w_i\right)^{\eta}$$

Zoning raises housing costs

- Supply: housing produced using fixed land and a variable input; zoning lowers productivity (e.g.,D'Amico et al. (2024))

$$H_i^S = \mathcal{A}_i e^{-(1+\eta)\kappa z_m} \left(q_i / w_i\right)^{\eta}$$

- Housing market clearing:

$$\ln q_i = \kappa z_m + \frac{1}{1+\eta}\left[\ln E_i - \ln\left(1 + t_m\right)\right] + \text{const}_i, \tag{1}$$

with local aggregate housing expenditure $E_i \equiv \sum_{\nu \in i} \beta_s y_\nu$

Zoning raises housing costs

- Supply: housing produced using fixed land and a variable input; zoning lowers productivity (e.g.,D'Amico et al. (2024))

$$H_i^S = \mathcal{A}_i e^{-(1+\eta)\kappa z_m} \left(q_i / w_i\right)^{\eta}$$

- Housing market clearing:

$$\ln q_i = \kappa z_m + \frac{1}{1+\eta}\left[\ln E_i - \ln\left(1 + t_m\right)\right] + \text{const}_i, \tag{1}$$

  with local aggregate housing expenditure $E_i \equiv \sum_{\nu \in i} \beta_s y_\nu$
- Resident-voters internalize the local rent response to zoning

Each type has an ideal policy

Type s values policy as

$$\ln V_s = \underbrace{\textstyle\sum_k \gamma_{sk} \ln \lambda_{mk}}_{\text{public goods mix}} + \underbrace{\Gamma_s \ln \tfrac{t_m}{1+t_m} - a\,\beta_s \ln(1+t_m)}_{\text{spending level vs. tax}} + \underbrace{\rho_s \ln z_m - \kappa \beta_s z_m}_{\text{amenity vs. rent}} + \underbrace{\Gamma_s \ln e_m}_{\text{tax base}}$$

given: public goods financing constraint and local rent response to zoning
$a = \eta/(1+\eta)$: residents' share of the tax; e_m: tax base per household

Each type has an ideal policy

Type s values policy as

$$\ln V_s = \underbrace{\textstyle\sum_k \gamma_{sk} \ln \lambda_{mk}}_{\text{public goods mix}} + \underbrace{\Gamma_s \ln \frac{t_m}{1+t_m} - a\,\beta_s \ln(1+t_m)}_{\text{spending level vs. tax}} + \underbrace{\rho_s \ln z_m - \kappa \beta_s z_m}_{\text{amenity vs. rent}} + \underbrace{\Gamma_s \ln e_m}_{\text{tax base}}$$

given: public goods financing constraint and local rent response to zoning
$a = \eta/(1+\eta)$: residents' share of the tax; e_m: tax base per household

Mix:	$\lambda_{sk} = \gamma_{sk}/\Gamma_s$	how a type values a differentiated public goods bundle
Tax:	$t_s^* = \dfrac{\Gamma_s}{a\,\beta_s}$	public goods value relative to the tax it bears, $a\beta_s$
Zoning:	$z_s^* = \dfrac{\rho_s}{\kappa\,\beta_s}$	amenity value relative to the rent it costs
Tax base:	non-satiating	more is always better

Local policy averages the ideal points

Probabilistic voting. Residents vote on utility differences up to an idiosyncratic taste.

Local policy averages the ideal points

Probabilistic voting. Residents vote on utility differences up to an idiosyncratic taste.

Platform convergence maximizes $\sum_s \mu_{ms} \ln V_s$ (Lindbeck and Weibull 1987):

$$t_m = \sum_s \omega_{ms}\, t_s^*, \qquad z_m = \sum_s \omega_{ms}\, z_s^*$$

$$\omega_{ms} = \frac{\mu_{ms}\beta_s}{\sum_{s'} \mu_{ms'}\beta_{s'}}, \qquad \lambda_{mk} = \frac{\sum_s \mu_{ms}\Gamma_s\, \lambda_{sk}}{\sum_{s'} \mu_{ms'}\Gamma_{s'}}$$

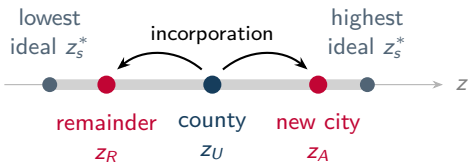
Local policy averages the ideal points

Probabilistic voting. Residents vote on utility differences up to an idiosyncratic taste.

Platform convergence maximizes $\sum_s \mu_{ms} \ln V_s$ (Lindbeck and Weibull 1987):

$$t_m = \sum_s \omega_{ms}\, t_s^*, \qquad z_m = \sum_s \omega_{ms}\, z_s^*$$

$$\omega_{ms} = \frac{\mu_{ms}\beta_s}{\sum_{s'} \mu_{ms'}\beta_{s'}}, \qquad \lambda_{mk} = \frac{\sum_s \mu_{ms}\Gamma_s\, \lambda_{sk}}{\sum_{s'} \mu_{ms'}\Gamma_{s'}}$$



Incorporation changes μ_{ms}
$\Rightarrow$ A's policies move toward its own residents
$\Rightarrow$ R's move toward its own residents too

Gains from incorporation

Resident of type s in $j \in \{A, R\}$ (incorporator or remainder; locations fixed):

$$\Delta \ln V_{js} = \underbrace{\textstyle\sum_k \gamma_{sk} \log \frac{\lambda_{jk}}{\lambda_{Uk}}}_{\text{public-goods mix}} + \underbrace{\Gamma_s \log \frac{t_j/(1+t_j)}{t_U/(1+t_U)} - a\beta_s \log \frac{1+t_j}{1+t_U}}_{\text{spending level and property tax}}$$
$$+ \underbrace{\Gamma_s \log \frac{e_j}{e_U}}_{\text{tax base per household}} + \underbrace{\rho_s \log \frac{z_j}{z_U} - \beta_s \kappa (z_j - z_U)}_{\text{zoning: amenity} - \text{housing cost}}$$

- **Policy fit:** each side sets public goods, tax and zoning for its own residents, so the average resident on each side gains

Gains from incorporation

Resident of type s in $j \in \{A, R\}$ (incorporator or remainder; locations fixed):

$$\Delta \ln V_{js} = \underbrace{\textstyle\sum_k \gamma_{sk} \log \frac{\lambda_{jk}}{\lambda_{Uk}}}_{\text{public-goods mix}} + \underbrace{\Gamma_s \log \frac{t_j/(1+t_j)}{t_U/(1+t_U)} - a\beta_s \log \frac{1+t_j}{1+t_U}}_{\text{spending level and property tax}}$$
$$+ \underbrace{\Gamma_s \log \frac{e_j}{e_U}}_{\text{tax base per household}} + \underbrace{\rho_s \log \frac{z_j}{z_U} - \beta_s \kappa (z_j - z_U)}_{\text{zoning: amenity} - \text{housing cost}}$$

- **Policy fit:** each side sets public goods, tax and zoning for its own residents, so the average resident on each side gains
- **Tax base:** a transfer: what the new city keeps, the remainder loses.

Incorporation: majority approval and a formation cost

- **Vote.** A resident votes yes if her gain exceeds the formation cost per household F_i, up to a voting taste (T1EV, scale $1/\zeta$):

$$S_i = \sum_s \mu_{is}\, \Lambda\big(\zeta\,[\Delta v_{is} - F_i]\big), \qquad \text{incorporate if } S_i \geq \tfrac{1}{2} \;\approx\; \Delta \bar{v}_i \geq F_i$$

Incorporation: majority approval and a formation cost

- **Vote.** A resident votes yes if her gain exceeds the formation cost per household F_i, up to a voting taste (T1EV, scale $1/\zeta$):

$$S_i = \sum_s \mu_{is}\, \Lambda\big(\zeta\,[\Delta v_{is} - F_i]\big), \qquad \text{incorporate if } S_i \geq \tfrac{1}{2} \;\approx\; \Delta \bar{v}_i \geq F_i$$

- **Formation cost**, a share of income per household: $F_i = \chi \ln L_i + h_{state} + \xi_i$
  falls with size L_i; state rules h_{state}; place shock ξ_i, logistic with scale $1/b$: organizers, local politics, the county's stance

Incorporation: majority approval and a formation cost

- **Vote.** A resident votes yes if her gain exceeds the formation cost per household F_i, up to a voting taste (T1EV, scale $1/\zeta$):

$$S_i = \sum_s \mu_{is}\, \Lambda\big(\zeta\,[\Delta v_{is} - F_i]\big), \qquad \text{incorporate if } S_i \geq \tfrac{1}{2} \;\approx\; \Delta\bar{v}_i \geq F_i$$

- **Formation cost**, a share of income per household: $F_i = \chi \ln L_i + h_{state} + \xi_i$
  falls with size L_i; state rules h_{state}; place shock ξ_i, logistic with scale $1/b$: organizers, local politics, the county's stance

- **Probability of incorporating:** $p_i = \Lambda\big(b\,(\Delta\bar{v}_i - \chi \ln L_i) + \psi_{state}\big)$

Incorporation: majority approval and a formation cost

- **Vote.** A resident votes yes if her gain exceeds the formation cost per household F_i, up to a voting taste (T1EV, scale $1/\zeta$):

$$S_i = \sum_s \mu_{is}\, \Lambda\big(\zeta\,[\Delta v_{is} - F_i]\big), \qquad \text{incorporate if } S_i \geq \tfrac{1}{2} \;\approx\; \Delta \bar{v}_i \geq F_i$$

- **Formation cost**, a share of income per household: $F_i = \chi \ln L_i + h_{state} + \xi_i$
  falls with size L_i; state rules h_{state}; place shock ξ_i, logistic with scale $1/b$: organizers, local politics, the county's stance

- **Probability of incorporating:** $p_i = \Lambda\big(b\,(\Delta \bar{v}_i - \chi \ln L_i) + \psi_{state}\big)$

Two votes

- given a map of jurisdictions, local policy voting sets equilibrium policy
- given the above policy rules, a referendum decides whether to redraw the map.

Roadmap

1. Motivating Facts
2. Model
3. Estimation
4. Gains from incorporation
5. Counterfactuals and policy implications

Estimation: preferences, cost of zoning, formation

Block	Parameter	Variation
Public goods	level Γ_s	Spending relative to residents' housing expenditure, within county, net of income (8,751 municipalities, COG 2017)
Public goods	mix $\lambda_{sk} = \gamma_{sk}/\Gamma_s$	Spending shares, within county
Zoning	cost κ	Quality-adjusted house prices on minimum lot size (6,541 municipalities)
Zoning	valuation ρ_s	Estimated minimum lot size, within county (8,010 municipalities)
Formation	b, χ	Which places incorporate, by modeled gain and size, within state and decade (21,742 place-decades)

Calibrated: housing shares β_s (CEX 2015–2019); $\eta = 2$; $\theta = 6$ (Bagagli 2025).

Untargeted moments: referendum votes, stated motives, the Fact 2 gradients, Facts 3 and 4.

Estimated preferences by resident type

	White college	Other college	White non-college	Other non-college
Public goods Γ_s	.020 (.007)	.029 (.019)	.025 (.006)	.034 (.021)
Preferred tax $t_s^* = \frac{\Gamma_s}{a\beta_s}$ (%)	15.4	21.8	16.5	19.8
Mix λ_{sk}: protection (%)	44.6 (3.6)	62.0 (4.5)	31.1 (1.9)	48.5 (2.3)
Mix λ_{sk}: infrastructure (%)	39.9 (3.4)	22.9 (4.6)	60.4 (2.1)	43.4 (2.4)
Mix λ_{sk}: amenities (%)	15.4 (1.5)	15.1 (2.0)	8.4 (0.7)	8.2 (1.1)
Zoning amenity ρ_s	**.049** (.005)	**.006** (.008)	**.044** (.004)	**.001** (.002)
Preferred stringency $z_s^* = \frac{\rho_s}{\kappa\beta_s}$	3.06	0.35	2.31	0.04
Preferred minimum lot (1,000 sq ft)	26.9	6.9	22.6	1.7
Housing share β_s	.193	.198	.230	.259

Cost of zoning: $\kappa = 0.067$ (0.010): a minimum lot 10% larger raises the price of housing by 1.0%. **Tests:** equal Γ_s not rejected ($p = 0.78$); equal mixes and equal ρ_s rejected. Two-step GMM; bootstrap SE (county resamples).

Estimation equations and details | Robustness

Formation: gains predict incorporation; the formation cost falls with size

1. Gains predict incorporation

$b = 49.8$ (8.5; clustered by state 16.7)
A gain of 0.1% of income raises the odds of incorporating by **5.1%**

1

How formation is estimated | Specifications | Four states

Formation: gains predict incorporation; the formation cost falls with size

1. Gains predict incorporation

$b = 49.8$ (8.5; clustered by state 16.7)
A gain of 0.1% of income raises the odds of incorporating by **5.1%**

2. The formation cost falls with size

$\chi = -0.756$ (0.146; clustered by state 0.346)
Doubling a place's households lowers its formation cost by **0.52%** of income

Households	Change in cost
1,000	0
2,000	−0.52
5,000	−1.22
20,000	−2.26

Against a place of 1,000 households; percent of income per household

1

How formation is estimated | Specifications | Four states

Formation: gains predict incorporation; the formation cost falls with size

1. Gains predict incorporation

$b = 49.8$ (8.5; clustered by state 16.7)
A gain of 0.1% of income raises the odds of incorporating by **5.1%**

2. The formation cost falls with size

$\chi = -0.756$ (0.146; clustered by state 0.346)
Doubling a place's households lowers its formation cost by **0.52%** of income

3. A place shock

SD 3.6% of income: what the modeled gain and size leave out

Households	Change in cost
1,000	0
2,000	−0.52
5,000	−1.22
20,000	−2.26

Against a place of 1,000 households; percent of income per household

1

How formation is estimated | Specifications | Four states

Untargeted moments: referendum votes and stated motives

Not used in estimation

1. Referendum votes

151 first referenda in 21 states: log odds of the yes share on the modeled average gain, with state effects

$\hat{\zeta} = 30.7$ (10.9): a gain of 1% of income moves a close vote by **7.7 pp**

California high-speed rail referendum: 5–16 pp (Fajgelbaum et al. 2023)

Untargeted moments: referendum votes and stated motives

Not used in estimation

1. Referendum votes

151 first referenda in 21 states: log odds of the yes share on the modeled average gain, with state effects

$\hat{\zeta} = 30.7$ (10.9): a gain of 1% of income moves a close vote by **7.7 pp**

California high-speed rail referendum: 5–16 pp (Fajgelbaum et al. 2023)

2. Stated motives

Share of incorporations in which the model tightens zoning

Proponents wanted to	Model tightens
restrict development	**70%** (44 of 63)
promote development	38% (6 of 16)

68% of the 79 cases match (Fisher $p = 0.022$)

Untargeted moments: who incorporates, data and model

Fact 2 regression, run on the data and on the model's probability of incorporating

pp of incorporating within a decade	Data	Model	Difference
Log income ratio (+10%)	0.098 (0.038)	0.095 (0.012)	0.003 (0.037)
College-share gap (+10 pp)	0.121 (0.146)	0.135 (0.035)	−0.014 (0.144)
Log population (doubling)	0.332 (0.074)	0.335 (0.036)	−0.003 (0.074)

The model matches the income and college gradients, which it does not target, and the size gradient, which χ fits.

Untargeted moments: who incorporates, data and model

Fact 2 regression, run on the data and on the model's probability of incorporating

pp of incorporating within a decade	Data	Model	Difference
Log income ratio (+10%)	0.098 (0.038)	0.095 (0.012)	0.003 (0.037)
College-share gap (+10 pp)	0.121 (0.146)	0.135 (0.035)	−0.014 (0.144)
Log population (doubling)	0.332 (0.074)	0.335 (0.036)	−0.003 (0.074)
White-share gap (+10 pp)	**0.116** (0.063)	−**0.057** (0.020)	**0.173** (0.061)

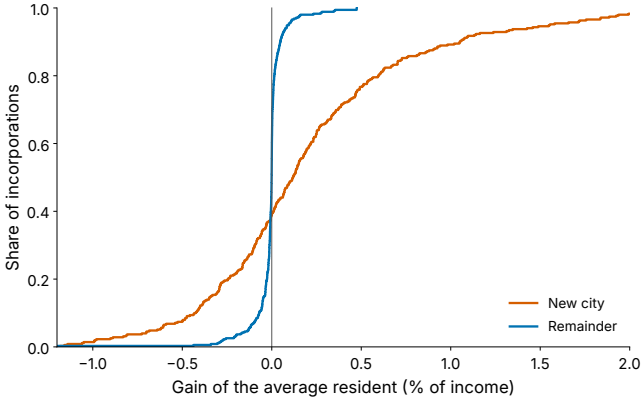
The model matches the income and college gradients, which it does not target, and the size gradient, which χ fits.

It does not produce the White-share gradient: preferences over policy, without preferences over neighbors, leave it unexplained.

Roadmap

1. Motivating Facts
2. Model
3. Estimation
4. Gains from incorporation
5. Counterfactuals and policy implications

Gains are small on average and vary widely

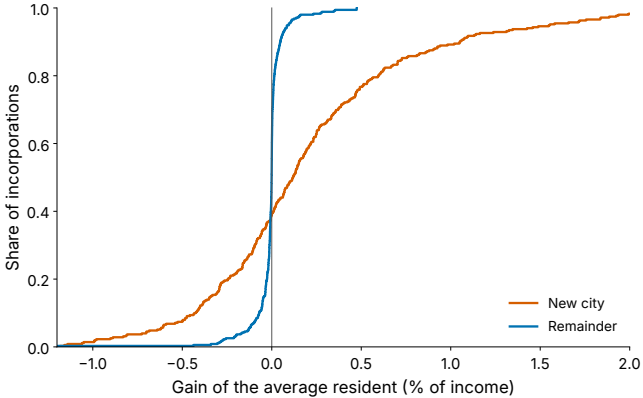


352 incorporations, residents' locations fixed

- Average resident of the incorporator: $+0.22\%$ of income

Residents move

Gains are small on average and vary widely

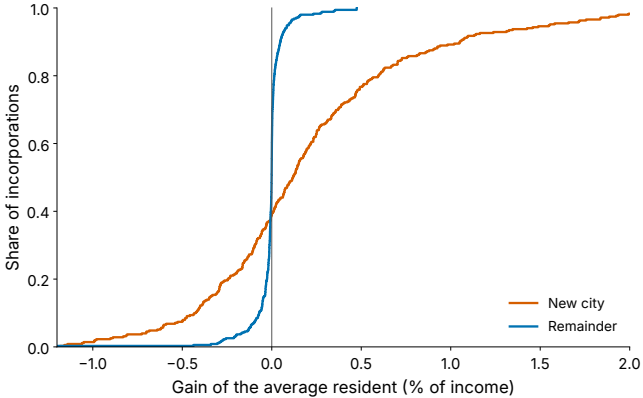


352 incorporations, residents' locations fixed

- Average resident of the incorporator: $+0.22\%$ of income
- Remainder: -0.013%

Residents move

Gains are small on average and vary widely

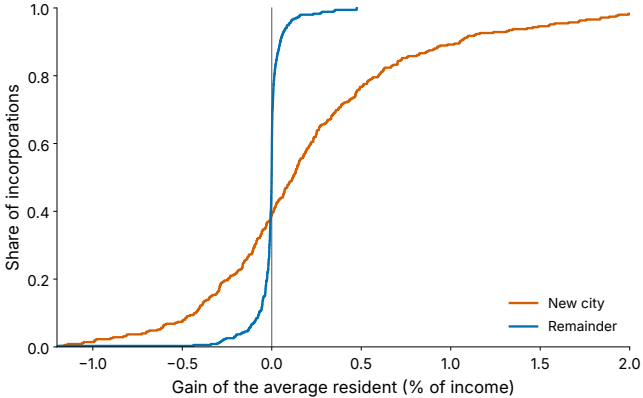


352 incorporations, residents' locations fixed

- Average resident of the incorporator: $+0.22\%$ of income
- Remainder: -0.013%
- Average resident loses in 39% of incorporators and 55% of remainders

Residents move

Gains are small on average and vary widely



352 incorporations, residents' locations fixed

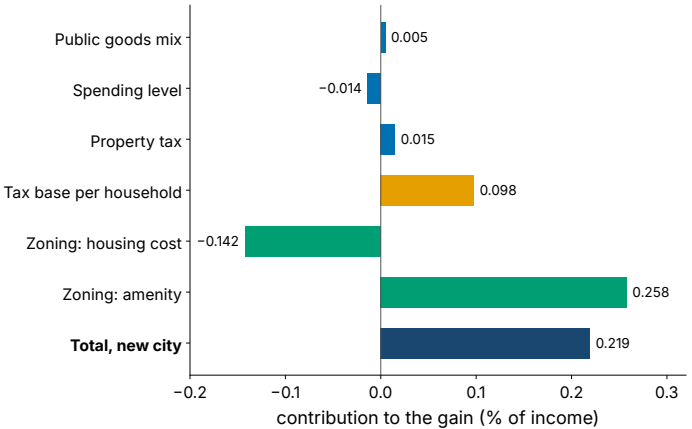
- Average resident of the incorporator: $+0.22\%$ of income
- Remainder: -0.013%
- Average resident loses in 39% of incorporators and 55% of remainders
- 10th to 90th pct across incorporators: -0.42% to $+1.05\%$

Residents move

Gains come from the tax base and stricter zoning, not public goods

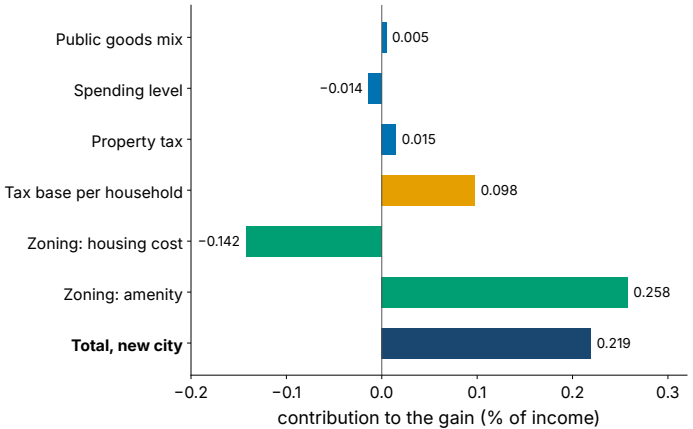
64% of new cities zone more strictly than their counties; 57% have a larger tax base per household

Gains come from the tax base and stricter zoning, not public goods



64% of new cities zone more strictly than their counties; 57% have a larger tax base per household

Gains come from the tax base and stricter zoning, not public goods



Remainder R (total -0.013)

Tax base	-0.015
Zoning: amenity	-0.003
Zoning: housing cost	$+0.004$
Public goods and tax	0.000

64% of new cities zone more strictly than their counties; 57% have a larger tax base per household

Gains go to residents whose preferred zoning prevails

% of income	New city			Remainder
	All residents	White majority (325)	Other majority (27)	residents
White college	**+0.61**	+0.28	−3.86	−0.026
Other college	−0.03	−0.05	+0.26	−0.002
White non-college	+0.15	+0.23	−3.33	−0.019
Other non-college	+0.03	−0.12	**+0.90**	+0.005

Gains go to residents whose preferred zoning prevails

% of income	New city			Remainder
	All residents	White majority (325)	Other majority (27)	residents
White college	**+0.61**	+0.28	−3.86	−0.026
Other college	−0.03	−0.05	+0.26	−0.002
White non-college	+0.15	+0.23	−3.33	−0.019
Other non-college	+0.03	−0.12	**+0.90**	+0.005

- White majority in 325 of 352 new cities: White residents gain through the zoning amenity; Other residents lose a little through housing costs

Gains go to residents whose preferred zoning prevails

% of income	New city			Remainder
	All residents	White majority (325)	Other majority (27)	residents
White college	**+0.61**	+0.28	−3.86	−0.026
Other college	−0.03	−0.05	+0.26	−0.002
White non-college	+0.15	+0.23	−3.33	−0.019
Other non-college	+0.03	−0.12	**+0.90**	+0.005

- White majority in 325 of 352 new cities: White residents gain through the zoning amenity; Other residents lose a little through housing costs
- Zoning loosens in 24 of the other 27: Other non-college residents gain through lower housing costs

Gains go to residents whose preferred zoning prevails

% of income	New city			Remainder
	All residents	White majority (325)	Other majority (27)	residents
White college	**+0.61**	+0.28	−3.86	−0.026
Other college	−0.03	−0.05	+0.26	−0.002
White non-college	+0.15	+0.23	−3.33	−0.019
Other non-college	+0.03	−0.12	**+0.90**	+0.005

- White majority in 325 of 352 new cities: White residents gain through the zoning amenity; Other residents lose a little through housing costs
- Zoning loosens in 24 of the other 27: Other non-college residents gain through lower housing costs
- Averaged over incorporations, every type's gain is slightly negative (−0.02 to −0.04%): each type loses heavily where it is a minority

Roadmap

1. Motivating Facts
2. Model
3. Estimation
4. Gains from incorporation
5. Counterfactuals and policy implications

Double consent stops affluent places

352 → 168

expected incorporations

−52%

Table

Double consent stops affluent places

352 → 168

expected incorporations

−52%

The remainder approves only if its average resident does not lose: it would veto 195 of the 352 observed incorporations.

Who is stopped: affluent places

	Stopped	Still occur
Income per household, place / remainder	**1.26**	0.85
Tax base per household, place / remainder	1.23	0.86
Households in the place	2,405	2,531
Gain, place's average resident (% of income)	+0.57	−0.16
Gain, remainder's average resident (% of income)	−0.04	+0.03

Size does not separate them: the rule halves expected incorporations in every size class.

Table

Compensation restores most incorporations

352 → 326

expected incorporations

−8%

double consent alone: 168

Table

Compensation restores most incorporations

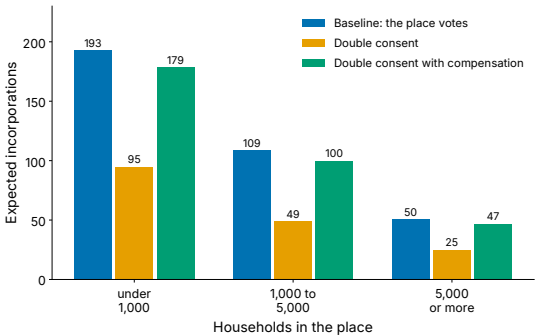
352 → 326

expected incorporations

−8%

double consent alone: 168

The new city pays the remainder's households enough to leave its average resident as well off as before



Of the 195 vetoed incorporations: payment a median 0.32% of the new city's income, against a median gain of 0.38%; all 195 keep a gain after paying

Table

California's revenue neutrality works more like a veto

Since 1992, a commission may approve an incorporation only if it is revenue neutral for the county: close to compensation, but paid to the county government, for lost revenue

California's revenue neutrality works more like a veto

Since 1992, a commission may approve an incorporation only if it is revenue neutral for the county: close to compensation, but paid to the county government, for lost revenue

Observed: remainder loses tax base

	1992 or earlier	1993 or later
California	18 of 31 (58%)	**2 of 13 (15%)**
Other states	32 of 55 (58%)	145 of 248 (58%)

California: Fisher $p = 0.018$

California's revenue neutrality works more like a veto

Since 1992, a commission may approve an incorporation only if it is revenue neutral for the county: close to compensation, but paid to the county government, for lost revenue

Observed: remainder loses tax base

	1992 or earlier	1993 or later
California	18 of 31 (58%)	**2 of 13 (15%)**
Other states	32 of 55 (58%)	145 of 248 (58%)

California: Fisher $p = 0.018$

Model: remainder loses tax base

share of expected California incorporations

Rule	Share
Baseline	55%
Revenue neutrality	49%
Double consent	0.3%

The drop after 1992 is larger than the model's rule predicts: closer to a veto, on 13 incorporations

Conclusion

Places form their own governments to keep their tax base and to zone more strictly; approval rules decide which places leave and who pays.

1. **Who incorporates:** larger, Whiter and richer places. Afterward, spending mixes and new construction diverge; population grows and the college share rises. Whiter places incorporate more than modeled gains predict.
2. **What they gain:** 0.22% of income: tax base 0.10%, zoning net of housing cost 0.12%, public goods almost nothing. Remainder −0.013%. Residents whose preferred zoning prevails gain.
3. **Rules decide who leaves and who pays:** double consent stops affluent places (−52%); with compensation, most still incorporate (−8%). California after 1992 works more like a veto.

Next steps

- **(Full) spatial equilibrium:** residents move and prices adjust across all places
- **Strategic interaction:** one place's incorporation changes its neighbors' gains; multiple equilibria
- **Homeowners:** gains from zoning and taxes capitalized into home values

Thank you!

jiesong@berkeley.edu

References

Alesina, Alberto, Reza Baqir, and Caroline Hoxby (2004). "Political jurisdictions in heterogeneous communities." In: *Journal of political Economy* 112.2, pp. 348–396.

Alesina, Alberto and Enrico Spolaore (1997). "On the number and size of nations." In: *The Quarterly Journal of Economics* 112.4, pp. 1027–1056.

Allen, Treb (2023). *The topography of nations*. Tech. rep. National Bureau of Economic Research.

Bagagli, Sara (2025). "The (Express)Way to Segregation: Evidence from Chicago." Working paper, July version. URL: https://drive.google.com/file/d/1f45RDOze8EiGvJVv1chgOp3cOPu5i_qn/view.

Bartik, Alexander, Arpit Gupta, and Daniel Milo (2025). "The Costs of Housing Regulation: Evidence from Generative Regulatory Measurement." SSRN 4627587.

Besley, Timothy and Stephen Coate (2003). "Centralized versus decentralized provision of local public goods: a political economy approach." In: *Journal of public economics* 87.12, pp. 2611–2637.

References

Bolton, Patrick and Gerard Roland (1997). "The breakup of nations: a political economy analysis." In: *The Quarterly Journal of Economics* 112.4, pp. 1057–1090.

Bordeu, Olivia (2025). "Commuting Infrastructure in Fragmented Cities." Working paper, June 25 version. URL: https://www.oliviabordeu.com/papers/fragmented_cities_obordeu.pdf.

Borusyak, Kirill, Xavier Jaravel, and Jann Spiess (2024). "Revisiting event-study designs: robust and efficient estimation." In: *Review of Economic Studies* 91.6, pp. 3253–3285.

Buchanan, James M. and Roger L. Faith (1987). "Secession and the Limits of Taxation: Toward a Theory of Internal Exit." In: *American Economic Review* 77.5, pp. 1023–1031.

Burns, Nancy (1994). *The formation of American local governments: Private values in public institutions*. New York: Oxford University Press.

Calabrese, Stephen M, Dennis N Epple, and Richard E Romano (2007). "On the Political Economy of Zoning." In: *Journal of Public Economics* 91.1–2, pp. 25–49.

— (2012). "Inefficiencies from metropolitan political and fiscal decentralization: Failures of Tiebout competition." In: *The Review of Economic Studies* 79.3, pp. 1081–1111.

References

D'Amico, Leonardo et al. (2024). *Why has construction productivity stagnated? The role of land-use regulation*. Tech. rep. National Bureau of Economic Research.

Dahis, Ricardo and Christiane Szerman (2026). "Decentralizing Development: The Economic Impacts of Government Splits." In: *American Economic Journal: Applied Economics* 18.3, pp. 220–253. DOI: 10.1257/app.20240009.

Desmet, Klaus et al. (2011). "The stability and breakup of nations: a quantitative analysis." In: *Journal of economic growth* 16.3, pp. 183–213.

Duranton, Gilles and Diego Puga (2023). "Urban Growth and Its Aggregate Implications." In: *Econometrica* 91.6, pp. 2219–2259. DOI: 10.3982/ECTA17936.

Fajgelbaum, Pablo D et al. (2019). "State taxes and spatial misallocation." In: *The Review of Economic Studies* 86.1, pp. 333–376.

Fajgelbaum, Pablo D et al. (2023). "Political Preferences and the Spatial Distribution of Infrastructure: Evidence from California's High-Speed Rail." In: *National Bureau of Economic Research*.

References

Fischel, William A. (2001). *The Homevoter Hypothesis: How Home Values Influence Local Government Taxation, School Finance, and Land-Use Policies*. Cambridge, MA: Harvard University Press.

Gallagher, Ryan, Allison Shertzer, and Tate Twinam (2026). *Zoning and the American Suburb*. Working Paper 26-37. Federal Reserve Bank of Philadelphia.

Gordon, Nora and Brian Knight (2009). "A spatial merger estimator with an application to school district consolidation." In: *Journal of Public Economics* 93.5-6, pp. 752–765.

Gravel, Nicolas and Rémy Oddou (2014). "The Segregative Properties of Endogenous Jurisdiction Formation with a Land Market." In: *Journal of Public Economics* 117, pp. 15–27. DOI: 10.1016/j.jpubeco.2014.02.001.

Hamilton, Bruce W. (1975). "Zoning and Property Taxation in a System of Local Governments." In: *Urban Studies* 12.2, pp. 205–211. DOI: 10.1080/00420987520080301.

Henderson, J. Vernon and Jacques-François Thisse (2001). "On Strategic Community Development." In: *Journal of Political Economy* 109.3, pp. 546–569.

References

Hilber, Christian A. L. and Frédéric Robert-Nicoud (2013). "On the Origins of Land Use Regulations: Theory and Evidence from US Metro Areas." In: *Journal of Urban Economics* 75, pp. 29–43. DOI: 10.1016/j.jue.2012.10.002.

Hochman, Oded, David Pines, and Jacques-François Thisse (1995). "On the Optimal Structure of Local Governments." In: *American Economic Review* 85.5, pp. 1224–1240.

Hsieh, Chang-Tai and Enrico Moretti (2019). "Housing Constraints and Spatial Misallocation." In: *American Economic Journal: Macroeconomics* 11.2, pp. 1–39. DOI: 10.1257/mac.20170388.

Jannin, Nicolas and Aurélie Sotura (2020). *This Town Ain't Big Enough? Quantifying Public Good Spillovers*. Working Paper. Banque de France.

Jehiel, Philippe and Suzanne Scotchmer (2001). "Constitutional Rules of Exclusion in Jurisdiction Formation." In: *Review of Economic Studies* 68.2, pp. 393–413.

Kenny, Lawrence W. and Adam Reinke (2011). "The Role of Income in the Formation of New Cities." In: *Public Choice* 149, pp. 75–88.

References

Konishi, Hideo (2008). "Tiebout's Tale in Spatial Economies: Entrepreneurship, Self-Selection, and Efficiency." In: *Regional Science and Urban Economics* 38, pp. 461–477.

Lindbeck, Assar and Jörgen W Weibull (1987). "Balanced-budget redistribution as the outcome of political competition." In: *Public choice* 52.3, pp. 273–297.

Macek, James (2026). "Housing regulation and neighborhood sorting across the United States." In: *Available at SSRN 7035859*.

McCasland, Jamie et al. (2026). "Shattered Metropolis: The Great Migration and the Fragmentation of Political Jurisdictions." In: *American Economic Journal: Economic Policy*. Conditionally accepted.

Miller, Gary J. (1981). *Cities by Contract: The Politics of Municipal Incorporation*. MIT Press. URL: https://mitpress.mit.edu/9780262131643/cities-by-contract/.

Musso, Juliet Ann (2001). "The political economy of city formation in California: Limits to Tiebout sorting." In: *Social Science Quarterly* 82.1, pp. 139–153.

References

Narasimhan, Veda and Jeffrey Weaver (2024). "Polity size and local government performance: evidence from India." In: *American Economic Review* 114.11, pp. 3385–3426.

Oates, Wallace E. (1999). "An Essay on Fiscal Federalism." In: *Journal of Economic Literature* 37.3, pp. 1120–1149.

Parkhomenko, Andrii (2023). "Local Causes and Aggregate Implications of Land Use Regulation." In: *Journal of Urban Economics* 138, p. 103605. DOI: 10.1016/j.jue.2023.103605.

Rollet, Vincent (2025). "Zoning and the Dynamics of Urban Redevelopment." Working paper, August 2025 version. URL: https://economics.mit.edu/people/phd-students/vincent-rollet.

Ruggieri, Francesco (2025). "Overlapping Jurisdictions and the Provision of Local Public Goods in U.S. Metropolitan Areas." Working paper, October 31 version.

Song, Jaehee (2025). "The Effects of Residential Zoning in US Housing Markets." In: *Journal of Urban Economics* 149, p. 103784. DOI: 10.1016/j.jue.2025.103784.

References

Turner, Matthew A, Andrew Haughwout, and Wilbert Van Der Klaauw (2014). "Land use regulation and welfare." In: *Econometrica* 82.4, pp. 1341–1403.

Weese, Eric (2015). "Political mergers as coalition formation: An analysis of the Heisei municipal amalgamations." In: *Quantitative Economics* 6.2, pp. 257–307.

Gains when residents move

Mean gain, average resident of the new city (% of income)	Benchmark	Robustness
Residents fixed	0.219	0.173
Residents move, $\theta = 6$	0.151	0.124
Residents move, $\theta = 3$	0.180	0.144

- Households move into or out of the new city (Fréchet, θ) and its housing price adjusts; the remainder and the rest of the county stay at their values at incorporation; first-order approximation
- With $\theta = 6$, households in the new city rise by 0.9%; the higher housing price costs its incumbents 0.06–0.08% of income, by type
- The median household gains in 222 of the 352 new cities, with and without moves

Back

What it takes to incorporate in four states

	California	Washington	Texas	Missouri
Incorporations in the sample	46	16	44	23
Statutory minimum	500 voters	1,500 residents[a]	201 residents[b]	none
Review before the vote	commission[c]	review board	none	none[d]
Median years to incorporate	2.05	2.14	0.44	0.81
(dated campaigns)	(25)	(14)	(8)	(3)
Attempts per incorporation	1.56	2.12	1.11	1.00
Median households	6,840	7,625	159	39
Smallest, households	1,157	1,156	23	2
(place, year)	(Loomis, 1984)	(Mill Creek, 1983)	(DISH, 2000)	(Three Creeks, 2008)

California counts registered voters. (a) 300 before 1994; higher near large cities. (b) 600 for a Type A city. (c) The Local Agency Formation Commission, with a fiscal analysis and, since 1992, a finding of revenue neutrality. (d) In St. Louis County, a boundary commission. Attempts per incorporation count all recorded attempts at the place up to its incorporation. Households at the census that opens the decade of incorporation.

Back

Approval rules: expected incorporations and who is stopped

	Observed	Baseline	Double consent	Double consent with compensation	Revenue neutrality
All	352	352.0	168.4	325.5	325.3
California	45	45.0	20.5	40.0	40.0
Fewer than 1,000 households	186	192.9	94.6	178.6	178.4
1,000 to 5,000	86	108.6	49.0	99.9	99.9
5,000 or more	80	50.5	24.9	47.0	47.0

Observed incorporations that the remainder vetoes: 195 of 352 (California 21 of 45). Payment, % of the new city's income: p10 0.06, median 0.32, p90 0.83, mean 0.40; median gain before payment 0.38%; all 195 keep a gain after paying.

Sums of probabilities over the 21,742 place-decades of the formation panel, with the estimates and state-by-decade effects at their baseline values. Double consent: the remainder approves if its average resident does not lose. Compensation: the new city pays the remainder's loss, in proportion to income. Revenue neutrality: single consent, and the new city pays the remainder's loss of tax base.

Who is stopped | Double consent | Compensation

Approval rules: who is stopped and who is restored

Incorporations weighted by their baseline probability

	Stopped by double consent	Kept under double consent	Restored by compensation
Expected incorporations	183.6	168.4	157.1
Income per household, place / remainder	1.26	0.85	1.23
Tax base per household, place / remainder	1.23	0.86	1.20
Households	2,405	2,531	2,427
Gain of the place's average resident (%)	0.57	−0.16	0.51
Gain of the remainder's average resident (%)	−0.04	0.03	−0.04
Zoning gap $\lvert \ln(z_A/z_R) \rvert$	0.19	0.33	0.19

Stopped: baseline minus double consent; restored: double consent with compensation minus double consent. Estimates of the formation logit, column (1).

Expected incorporations | Double consent | Compensation

Preferences: robustness set and tests

Robustness: log population added to every block	White college	Other college	White non-college	Other non-college
Public goods Γ_s	.021 (.006)	.032 (.010)	.024 (.003)	.035 (.014)
Mix: protection (%)	41.2 (2.1)	48.9 (3.1)	35.1 (1.2)	42.8 (1.6)
infrastructure (%)	44.9 (2.1)	41.4 (3.1)	54.8 (1.2)	51.2 (1.6)
amenities (%)	13.9 (1.2)	9.7 (1.7)	10.1 (0.6)	5.9 (0.9)
Zoning amenity ρ_s	.068 (.005)	.054 (.009)	.026 (.004)	.012 (.003)

Tests	Benchmark	Robustness
Γ_s: J (df 3) [p]	6.64 [0.084]	6.26 [0.100]
Γ_s equal across types: Wald (df 2) [p]	0.49 [0.782]	0.67 [0.714]
λ_{sk}: J (df 2) [p]	9.89 [0.007]	12.36 [0.002]
λ_{sk} equal across types: Wald (df 4)	105.6	61.3
ρ_s: J (df 3) [p]	31.14 [<0.001]	0.29 [0.962]
ρ_s equal across types: Wald (df 2)	149.8	104.1

With the robustness set, the average gain is 0.173% of income: tax base 0.099%, zoning net of housing cost 0.070%; averaged over incorporations, the two college types gain 0.160% and 0.157%, the non-college types 0.029% and 0.010%. Under additivity, a parameter differs across types only through a race difference and an education difference; the Wald tests set both to zero.

Back

Formation estimates across specifications

	(1) Main	(2) Main, robust prefs	(3) Conditional logit	(4) State, decade	(5) Decade	(6) Constant, 32 states	(7) Constant, all
b	49.80	62.33	49.17	50.56	45.71	40.74	44.32
model SE	(8.45)	(9.22)	(8.38)	(8.44)	(7.76)	(7.46)	(7.57)
clustered SE	[16.67]	[14.18]	[16.39]	[16.62]	[13.78]	[13.31]	[13.20]
χ (% of income per unit of ln L)	−0.756	−0.588	−0.755	−0.696	−0.397	−0.407	−0.378
model SE	(0.146)	(0.106)	(0.146)	(0.134)	(0.095)	(0.105)	(0.094)
clustered SE	[0.346]	[0.283]	[0.345]	[0.335]	[0.291]	[0.343]	[0.310]
Doubling L lowers the cost by (%)	0.524	0.407	0.523	0.482	0.275	0.282	0.262
SD of the place shock (%)	3.64	2.91	3.69	3.59	3.97	4.45	4.09
Place-decades	12,557	12,557	12,557	18,568	21,742	18,568	21,742
States (clusters)	32	32	32	32	42	32	42

Logit of incorporation within a decade on the average household's gain and ln L; the coefficient on ln L is $-b\chi$. (1): one fixed effect per state-by-decade cell; cells without an incorporation drop out. (2): the robustness set's gains. (3): conditional logit within the same cells. (4)–(7): state and decade dummies; decade dummies; a constant. Model SE in parentheses, clustered by state in brackets. Place shock SD $= \pi/(\sqrt{3}\,b)$. Counterfactuals with the other specifications: double consent 167.9–174.4, with compensation 324.4–328.2.

Back

Closest papers

	Endogenous formation	Mobility	Quantified	Land use policies
Jurisdiction formation				
Alesina–Spolaore (1997); Bolton–Roland (1997)	✓			
Gravel–Oddou (2014)	✓	✓		
Konishi (2008)	✓	✓		✓
Desmet et al. (2011); Weese (2015)	✓		✓	
Decentralization and fragmented governments				
Calabrese et al. (2012)		✓	✓	
Jannin–Sotura (2020); Bordeu (2025); Ruggieri (2025)		✓	✓	
land use regulation				
Hsieh–Moretti (2019); Macek (2026)		✓	✓	
Parkhomenko (2023)		✓	✓	✓
This paper	✓	✓†	✓	✓

Back

Related literature by strand (1/2)

Jurisdiction formation

- Theory: Buchanan and Faith (1987), Alesina and Spolaore (1997), Bolton and Roland (1997), and Jehiel and Scotchmer (2001)
- With a land market, stylized: Hochman et al. (1995), Henderson and Thisse (2001), Konishi (2008), and Gravel and Oddou (2014)
- Quantitative: Gordon and Knight (2009), Desmet et al. (2011), Weese (2015), and Allen (2023)
- US evidence: Burns (1994), Musso (2001), Alesina et al. (2004), Kenny and Reinke (2011), and McCasland et al. (2026)

Back

Related literature by strand (2/2)

Decentralization

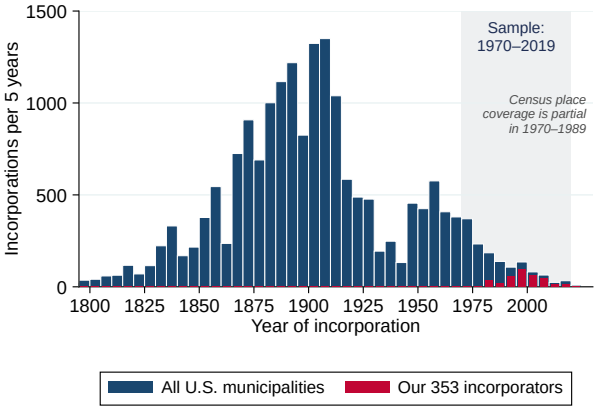
- Theory: Oates (1999) and Besley and Coate (2003)
- Quantitative, map given: Calabrese et al. (2012), Fajgelbaum et al. (2019), Jannin and Sotura (2020), Bordeu (2025), and Ruggieri (2025)
- Splits and polity size: Narasimhan and Weaver (2024) and Dahis and Szerman (2026)

land use regulation

- Political economy: Hamilton (1975), Fischel (2001), Calabrese et al. (2007), Hilber and Robert-Nicoud (2013), Parkhomenko (2023), and Duranton and Puga (2023)
- Effects, regulation given: Turner et al. (2014), Hsieh and Moretti (2019), Song (2025), and Macek (2026)
- Zoning and incorporation: Gallagher et al. (2026)

Back

Municipal incorporation in the U.S.



My 353 incorporations

24% of the 1,370 U.S. incorporations dated 1970–2019 are in the sample

1980s: 17%; 1990s: 58%; 2000s: 67%; 2010s: 49%

Census place coverage is partial in 1970–1989.

Back

Geography of Recent Incorporation

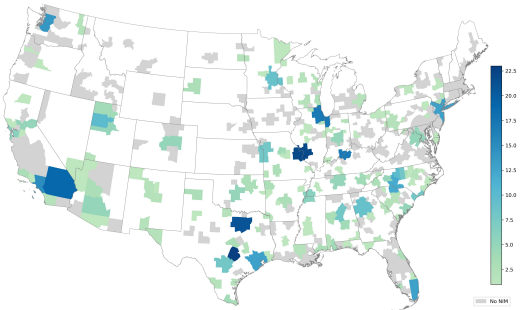


Figure 1: Number of recent incorporation

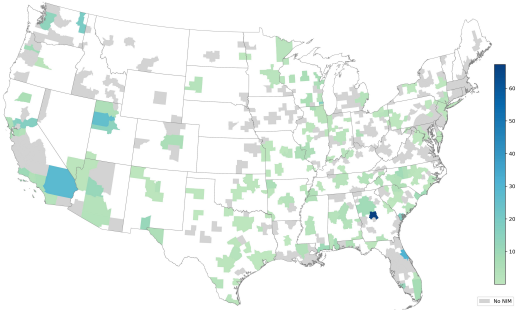


Figure 2: Percentage of metropolitan population in recent incorporators

Data

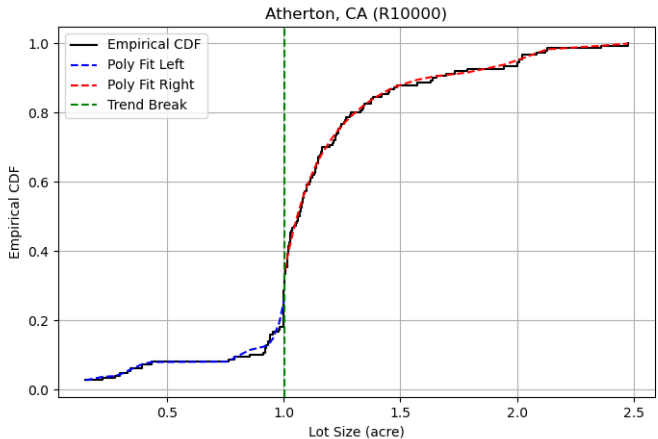


Figure 3: Estimating Minimum Lot Size (MLS) in Atherton, CA (R1) following Song (2025)

Fact 1: local policies differ with constituency

	(1) Log MLS	(2) Log exp. per HH	(3) Log tax base	(4) Log tax rate	(5) Log exp. per HH	(6) Public safety	(7) Infra-structure
White share (per 10 pp)	9.11	−8.67	3.39	−12.06	−2.93	−1.02	0.54
	(0.57)	(1.05)	(0.22)	(1.10)	(0.92)	(0.22)	(0.22)
College share (per 10 pp)	4.74	13.80	14.57	−0.76	7.31	0.30	−0.84
	(0.86)	(0.98)	(0.32)	(1.07)	(1.77)	(0.33)	(0.34)
Log population, log income					yes	yes	yes
Mean of outcome	9.47	7.15	9.56	−2.41	7.15	38.5	51.5
Municipalities	7,998	8,601	8,601	8,601	8,601	8,601	8,601
Counties	948	971	971	971	971	971	971

County effects; SE clustered by county. Columns (1) to (5): log points per 10 pp of the share; (6) and (7): pp of spending on the three categories. Column (2) equals (3) plus (4): log spending per household is the log tax base (residents' housing expenditure per household) plus the log tax rate, $t/(1+t)$. Both shares enter every regression.

Zoning Spending Mix

Fact 2: who incorporates

pp of incorporating within a decade	(1) Linear	(2) Quadratic
White-share gap Δ^W (+10 pp)	0.114 (0.062)	0.204 (0.090)
College-share gap Δ^C (+10 pp)	0.131 (0.145)	0.082 (0.129)
Log income ratio (+10%)	0.098 (0.038)	0.109 (0.039)
Log population (doubling)	0.333 (0.073)	0.339 (0.075)
$(\Delta^W)^2$		0.021 (0.017)
$(\Delta^C)^2$		0.007 (0.058)
$\Delta^W \Delta^C$		0.071 (0.083)
Average marginal effect of Δ^W		0.197 (0.086)
Average marginal effect of Δ^C		0.078 (0.129)
Quadratic terms, $F(3, 873)$ $[p]$		1.21 [0.305]
Eigenvalues of the quadratic form		0.050 (0.039); −0.022 (0.064)
Mean of the outcome (pp)	1.621	1.621
Place-decades; places; incorporations	21,774; 6,683; 353	
State-by-decade cells; counties	175; 874	

Linear probability models with state-by-decade effects; SE clustered by county. Gaps: the place minus its unincorporated county (in township states, the township), a decade earlier. In column (2) the linear terms are slopes at zero gaps. The panel excludes 436 place-decades in which a dated state rule ruled out incorporation.

Back

Stated motives of incorporators

Motives that proponents stated, from the attempt that succeeded: a record for 284 of the 353 incorporations

Proponents sought to	Incorporations	Share of the 353
get better or different public goods	125	35%
restrict development	78	22%
control land use in other ways	98	28%
avoid annexation by a neighboring city	129	37%
stop subsidizing other areas	24	7%

A place can state several motives. Coded from newspaper reports, official documents and local histories. The 69 incorporations without a record are mostly small places.

Back

What follows incorporation: incorporators vs. failed attempts

	Log population	Log households	Log housing units	Log home value	White share (pp)	College share (pp)
Average, decades 0 to 2	0.210	0.210	0.192	0.028	0.54	2.69
	(0.048)	(0.049)	(0.050)	(0.028)	(0.81)	(0.60)
Joint test of leads (p)	0.24	0.40	0.14	0.09	0.28	0.10

	White college share (pp)	Other college share (pp)	White non-college (pp)	Other non-college (pp)
Average, decades 0 to 2	1.43 (0.55)	1.20 (0.43)	−1.30 (0.60)	−1.33 (0.60)
Joint test of leads (p)	0.50	0.78	0.41	0.09

Imputation estimator (Borusyak et al. 2024) with place effects and state-by-decade effects; SE clustered by state. 326 incorporations with outcomes after incorporation (318 for the four type shares, which start in 1980), in 38 states, against 141 places whose formal attempt to incorporate failed and that had not incorporated by 2020. With baseline controls interacted with decade: log population 0.20 (0.05), college share 1.8 pp (0.6).

Back

Places that incorporate and places that do not

	Fact 2 panel		Fact 4 panel, baseline	
	Incorporating place-decades	Never incorporating	Incorporators	Failed attempts
Households	3,592 (6,282)	1,666 (3,541)	2,442 (4,072)	3,608 (4,861)
Population	9,915 (17,149)	4,655 (9,531)	7,386 (12,371)	10,391 (14,221)
Non-Hispanic White, adults (%)	85.6 (21.4)	81.1 (25.5)	89.0 (18.6)	89.0 (13.9)
College, adults (%)	22.2 (14.8)	21.4 (15.5)	18.0 (12.3)	21.6 (14.8)
Median household income (2022 $)	87,655 (32,556)	80,953 (33,583)	79,615 (26,628)	83,379 (34,227)
White-share gap with U (pp)	2.7 (15.1)	−0.4 (17.2)	1.9 (13.6)	3.0 (12.3)
College-share gap with U (pp)	2.0 (10.8)	−0.9 (10.2)	1.0 (9.8)	2.8 (11.1)
Log income ratio with U	0.058 (0.243)	−0.028 (0.259)	0.044 (0.236)	0.041 (0.275)
Observations	353	21,166	353	141

Means, standard deviations in parentheses. Fact 2 panel: the 353 place-decades in which a place incorporates and the 21,166 place-decades of places that never incorporate. Fact 4 panel: the 353 incorporators and the 141 places whose attempts to incorporate failed, each at its first baseline year; 62 incorporators also had an earlier failed attempt. Gaps: the place minus its unincorporated county U. Median household income excludes the 1970 census.

Back

Samples

Analysis	Unit	Period	Observations	Incorporations
Fact 1, zoning block	municipality	2018–2022	7,998	
Fact 1, fiscal block	municipality	2017	8,601	
Cost of zoning (κ)	municipality	2010–2022	6,541	
Fact 2 and formation	place-decade	1970–2020	21,742	352
Referenda	first referendum	1970–2020	151	
Fact 3, spending mix	incorporator	1977–2017		186
Fact 3, new lots	incorporator	1970–2020		195
Fact 4	place-decade	1970–2020	141 failed places	326
Gains and approval rules	incorporator	1970–2020		352

Municipalities are counted after dropping counties with a single municipality from regressions with county effects: 948 (zoning), 971 (public goods) and 799 (cost of zoning) counties. The preferences over public goods are estimated on 8,751 municipalities in 1,121 counties. Fact 2 excludes place-decades in which a dated state rule made incorporation impossible; Dublin, California (1982), lacks the inputs of the tax-base gap.

Back

Estimation: preferences

Public goods: level Γ_s (county effects, log income per household)

A municipality spends the share $\bar{\Gamma}_m/(\bar{\Gamma}_m + a\bar{\beta}_m)$ of its residents' housing expenditure e_m:

$$\frac{Q_m}{L_m} = \frac{\bar{\Gamma}_m}{\bar{\Gamma}_m + a\bar{\beta}_m} e_m + \alpha_{n(m)} + \alpha_y \ln \frac{Y_m}{L_m} + u_m$$

Q_m: spending on the three categories;
$e_m = (\mu_m'\beta)Y_m/L_m$. Moments: residual orthogonal to three type shares and the squared White and college shares, within county; mean spending matched. Additive: $\Gamma_1 + \Gamma_4 = \Gamma_2 + \Gamma_3$.

Public goods: mix λ_{sk} (county effects)

$\lambda_{mk} = \sum_s \tilde{\mu}_{ms}\lambda_{sk}$, $\tilde{\mu}_{ms} = \mu_{ms}\Gamma_s/\bar{\Gamma}_m$, for protection and infrastructure; mean shares matched; additive.

Zoning: cost κ

Log price of a reference home (hedonic, block-group-by-jurisdiction effects) on log MLS, with county effects, log E_m and $\log(1 + t_m)$: elasticity 0.106 (0.016), so $\kappa = 0.106\, d = 0.067$ (0.010); 6,541 municipalities.

Zoning: valuation ρ_s (county effects)

$z_m = \exp\{(\log \mathrm{MLS}_m - 9.498)/d\}$, $d = 0.628$: mean and SD of log MLS across municipalities.

$$z_m\bar{\beta}_m = \sum_s \mu_{ms}\rho_s/\kappa + \alpha_{n(m)} + u_m$$

Same moments; the equation holds at the sample mean; $\rho_s \geq 0$; additive. 8,010 municipalities.

Inference

Two-step GMM. Bootstrap of counties that repeats every step: 199 draws (public goods), 99 draws of linked county groups (zoning).

Back

Estimation: formation

Model

1. Vote. Yes share $S_i(F) = \sum_s \mu_{is} \Lambda(\zeta[\Delta v_{is} - F])$. The cost at which $S_i = \frac{1}{2}$ is about the average gain $\Delta\bar{v}_i$: near zero Λ is close to linear, and the third central moment of the types' gains moves the threshold only a little.

2. Cost per household: $F_i = \chi \ln L_i + h_{state} + \xi_i$, ξ_i logistic with scale $1/b$: a shock to the whole place.

3. Incorporate if $\Delta\bar{v}_i \geq F_i$:
$p_i = \Lambda\Big(b(\Delta\bar{v}_i - \chi \ln L_i) + \psi_{state}\Big)$, $\psi_{state} = -b\, h_{state}$.

Why the place shock. Voting tastes average out over a place's voters; without ξ_i a place with a given gain and size would incorporate for sure or never.

Estimation

Logit of incorporation within a decade on $\Delta\bar{v}_{it}$ and $\ln L_{it}$, both from the decade before, with state-by-decade effects $\psi_{state \times t}$; the coefficient on $\ln L$ is $-b\chi$.

Identification: b from how much more often places with larger modeled gains incorporate than others in the same state and decade; χ from how much more often larger places incorporate at the same gain.

82 cells with at least one incorporation and one place that does not: 12,557 place-decades in 32 states. The fixed effects match expected to observed incorporations in each cell.

Not used: the 151 referenda give $\hat{\zeta} = 30.7$ (10.9) separately; with $\ln L$ added, $\chi_{vote} = 0.50$ (0.38), the opposite sign.

Specifications Back

Backup from the 1016 deck

Not updated to the 1103 draft: these slides use earlier samples, motive coding, specifications or model versions, and their numbers can differ from the main slides.

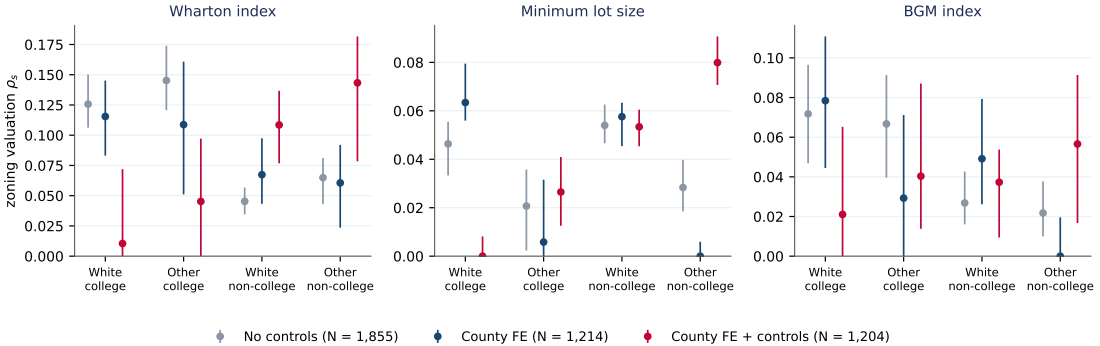
State legal frameworks for incorporation

	Incorporations 1980–2020	K_{state}	Population minimum	Petition signatures	External approval	Feasibility review	Vote or hearing	Legal score
California	39	95.4	500 electors	agency route	LAFCO	yes	yes	0.80 (5)
Wisconsin	12	88.1	575	50 electors	circuit court or board	–	yes	1.00 (4)
Washington	16	79.8	900	10%	boundary review board or county	–	yes	1.00 (4)
Georgia	9	75.3	200	–	not coded	–	–	(1)
Alabama	17	1.6	1,150	15%	county probate judge	–	yes	1.00 (4)
Illinois	22	1.1	1,350	200 electors	circuit court	–	yes	1.00 (4)
Texas	44	0.3	467	10%	county judge	–	yes	1.00 (4)
North Carolina	38	0.1	none	15%	General Assembly	yes	–	0.75 (4)
Utah	14	0.06	450	20%	county clerk and legislative body	–	–	1.00 (3)
Florida	21	≤ 0	1,500	no petition	state legislature (special act)	yes	–	0.75 (4)
Missouri	23	≤ 0	167	15%	county commission	–	yes	0.83 (4)
Kentucky	11	≤ 0	300	67%	circuit court	yes	–	1.00 (4)
New York	13	≤ 0	250	–	state legislature (cities)	–	–	(1)

K_{state}: estimated fixed cost in household incomes per year (logit vote share). Population minimum: residents unless stated, averaged over the ordinary municipal forms (Census 1990, Table 9-1). Legal score: mean of five domains (eligibility, initiation, external approval, feasibility, vote or hearing), observed domains in parentheses; – not stated in the sources, not absent. The minimum does not line up with the estimated costs (rank correlation -0.13 with the implied size threshold).

Back

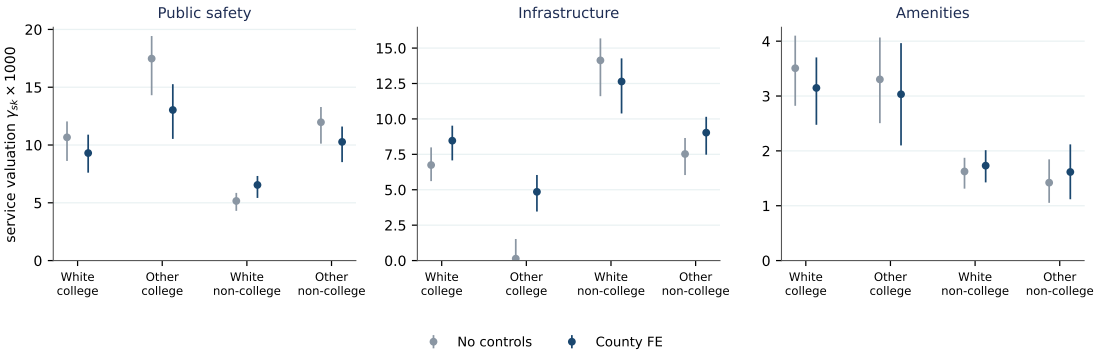
Zoning valuations by specification



With county fixed effects, college residents value zoning more on all three measures. Holding home value, income, population and density fixed reverses the ranking for the Wharton index and minimum lot size; the White college estimate is at zero in 31% (Wharton), 85% (lot size) and 6% (BGM) of bootstrap draws.

Additive restriction; 95% bootstrap intervals (99 linked-county resamples). Samples: Wharton 1,855 / 1,214 / 1,204 cities; minimum lot size 8,779 / 8,601 / 8,424; BGM 4,845 / 4,008 / 3,941.

Public goods valuations by specification



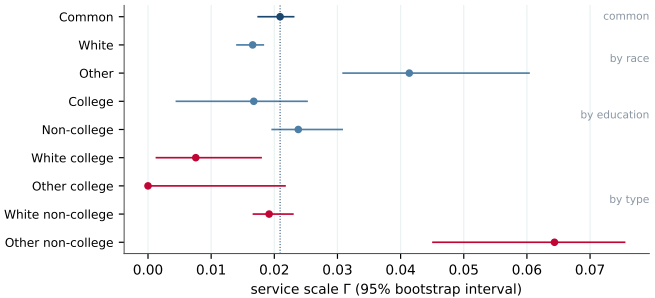
County fixed effects shrink the differences across types by about half: the race contrast in the safety share falls from 32.6 pp to 17.8 pp, the education contrast in the infrastructure share from 35.3 pp to 20.0 pp. Workload controls for public goods are not yet run.

$\gamma_{sk} = \Gamma\, \omega_{sk}$ with the common $\Gamma = .0209$; additive restriction; 95% intervals from 399 county resamples; 8,751 cities.

The scale of public goods valuations by type

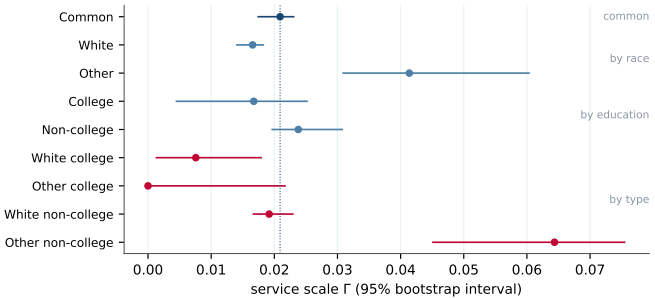
Spending-level equation, 7 moments, no county effects in any specification; 95% intervals from 399 county resamples; 8,751 cities.

The scale of public goods valuations by type



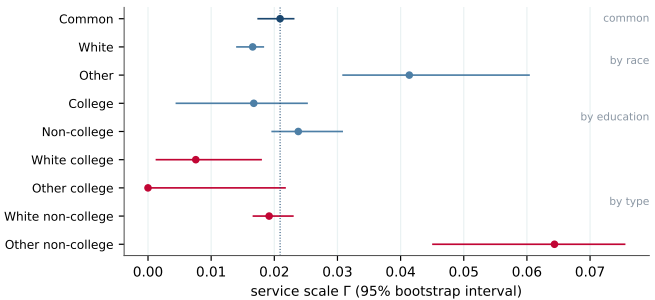
Spending-level equation, 7 moments, no county effects in any specification; 95% intervals from 399 county resamples; 8,751 cities.

The scale of public goods valuations by type



Spending-level equation, 7 moments, no county effects in any specification; 95% intervals from 399 county resamples; 8,751 cities.

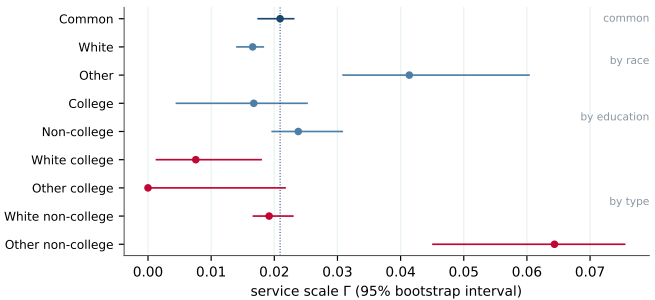
The scale of public goods valuations by type



- Common scale: $\Gamma = .0209$ (.0016)

Spending-level equation, 7 moments, no county effects in any specification; 95% intervals from 399 county resamples; 8,751 cities.

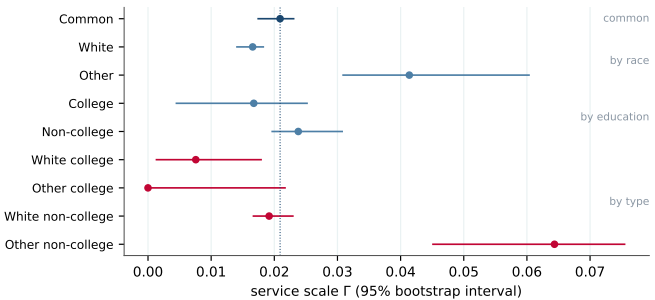
The scale of public goods valuations by type



- Common scale: $\Gamma = .0209$ (.0016)
- By race: other residents 2.5 times White residents

Spending-level equation, 7 moments, no county effects in any specification; 95% intervals from 399 county resamples; 8,751 cities.

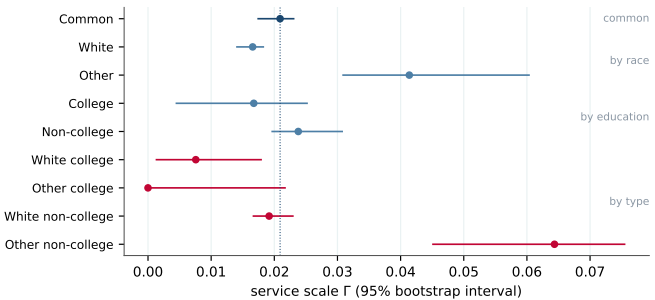
The scale of public goods valuations by type



- Common scale: $\Gamma = .0209$ (.0016)
- By race: other residents 2.5 times White residents
- By type: non-college above college; the other college estimate is at zero in 346 of 399 draws

Spending-level equation, 7 moments, no county effects in any specification; 95% intervals from 399 county resamples; 8,751 cities.

The scale of public goods valuations by type



- Common scale: $\Gamma = .0209$ (.0016)
- By race: other residents 2.5 times White residents
- By type: non-college above college; the other college estimate is at zero in 346 of 399 draws
- The main table and the gains use Γ_s with county effects, log population and log income: .021, .032, .024, .035 (equal values not rejected, $p = 0.20$)

Spending-level equation, 7 moments, no county effects in any specification; 95% intervals from 399 county resamples; 8,751 cities.

Restrict-motive cities build less, and so does their county

Restrict − none	Years 0–10	Years 10–20
Own new homes	−0.34 (0.13)	−0.58 (0.24)
County new homes	−0.32 (0.09)	−0.54 (0.21)
New city minus county	0.00 (0.12)	−0.04 (0.21)

Log points: change in log(1 + new single-family homes per decade) from the 10 years before; 58 to 62 restrict and 230 to 242 no-land use-motive cities with data, by cell; SE clustered by county.

Why measures relative to the county miss it:

- The county builds less too, so the gap does not move

Restrict-motive cities build less, and so does their county

Restrict − none	Years 0–10	Years 10–20
Own new homes	−0.34 (0.13)	−0.58 (0.24)
County new homes	−0.32 (0.09)	−0.54 (0.21)
New city minus county	0.00 (0.12)	−0.04 (0.21)

Log points: change in log(1 + new single-family homes per decade) from the 10 years before; 58 to 62 restrict and 230 to 242 no-land use-motive cities with data, by cell; SE clustered by county.

Why measures relative to the county miss it:

- The county builds less too, so the gap does not move
- Under the baseline coding, the lot change comes in the second decade

Restrict-motive cities build less, and so does their county

Restrict − none	Years 0–10	Years 10–20
Own new homes	−0.34 (0.13)	−0.58 (0.24)
County new homes	−0.32 (0.09)	−0.54 (0.21)
New city minus county	0.00 (0.12)	−0.04 (0.21)

Log points: change in log(1 + new single-family homes per decade) from the 10 years before; 58 to 62 restrict and 230 to 242 no-land use-motive cities with data, by cell; SE clustered by county.

Why measures relative to the county miss it:

- The county builds less too, so the gap does not move
- Under the baseline coding, the lot change comes in the second decade
- A minimum-lot detector needs 50 new lots: of the cities with 50 or more before, 26% of restrict cities fall below it by years 10–20, against 19% of cities with no land use motive (about one SE apart)

Restrict-motive cities build less, and so does their county

Restrict − none	Years 0–10	Years 10–20
Own new homes	−0.34 (0.13)	−0.58 (0.24)
County new homes	−0.32 (0.09)	−0.54 (0.21)
New city minus county	0.00 (0.12)	−0.04 (0.21)

Log points: change in log(1 + new single-family homes per decade) from the 10 years before; 58 to 62 restrict and 230 to 242 no-land use-motive cities with data, by cell; SE clustered by county.

Why measures relative to the county miss it:

- The county builds less too, so the gap does not move
- Under the baseline coding, the lot change comes in the second decade
- A minimum-lot detector needs 50 new lots: of the cities with 50 or more before, 26% of restrict cities fall below it by years 10–20, against 19% of cities with no land use motive (about one SE apart)
- 7 of the 63 oppose only a facility, which single-family lots cannot show

Restrict-motive cities build less, and so does their county

Restrict − none	Years 0–10	Years 10–20
Own new homes	−0.34 (0.13)	−0.58 (0.24)
County new homes	−0.32 (0.09)	−0.54 (0.21)
New city minus county	0.00 (0.12)	−0.04 (0.21)

Log points: change in log(1 + new single-family homes per decade) from the 10 years before; 58 to 62 restrict and 230 to 242 no-land use-motive cities with data, by cell; SE clustered by county.

Why measures relative to the county miss it:

- The county builds less too, so the gap does not move
- Under the baseline coding, the lot change comes in the second decade
- A minimum-lot detector needs 50 new lots: of the cities with 50 or more before, 26% of restrict cities fall below it by years 10–20, against 19% of cities with no land use motive (about one SE apart)
- 7 of the 63 oppose only a facility, which single-family lots cannot show
- Restrict cities were denser before incorporation: +0.56 log points of housing units per km^2 (SE 0.23)

A check on the zoning rule: predicted change and what cities did

The rule leans the right way on stated motives, with borderline precision; city by city it gets the direction of the lot change right 56% of the time.

ρ_s comes only from existing cities. The predicted change is the zoning rule $z = \mu \cdot \rho/(\kappa\,\mu \cdot \beta)$ at the new city's composition minus at the county's before incorporation. Current E grid parameters (provisional). Stated motives: column 2 coding, 60 restrict, 246 none, 16 promote among the 344 model incorporations. SE clustered by county.

A check on the zoning rule: predicted change and what cities did

Predicted change in zoning, $\log z_A - \log z_U$
(log points)

Stated motive	Mean	vs. none
Restrict	+0.042	+0.031 (0.016)
None	+0.011	
Promote	−0.053	−0.064 (0.026)

With decade effects and the number of sources:
+0.021 (0.017) and −0.058 (0.028).

The rule leans the right way on stated motives, with borderline precision; city by city it gets the direction of the lot change right 56% of the time.

ρ_s comes only from existing cities. The predicted change is the zoning rule $z = \mu \cdot \rho/(\kappa\,\mu \cdot \beta)$ at the new city's composition minus at the county's before incorporation. Current E grid parameters (provisional). Stated motives: column 2 coding, 60 restrict, 246 none, 16 promote among the 344 model incorporations. SE clustered by county.

A check on the zoning rule: predicted change and what cities did

Predicted change in zoning, $\log z_A - \log z_U$ (log points)

Stated motive	Mean	vs. none
Restrict	+0.042	+0.031 (0.016)
None	+0.011	
Promote	−0.053	−0.064 (0.026)

With decade effects and the number of sources: +0.021 (0.017) and −0.058 (0.028).

Observed change per +0.1 predicted

	Years 0–10	10–20
Median lot gap	+0.11 (0.05)	+0.21 (0.09)
Own new homes	−0.16 (0.07)	−0.33 (0.12)

The rule leans the right way on stated motives, with borderline precision; city by city it gets the direction of the lot change right 56% of the time.

ρ_s comes only from existing cities. The predicted change is the zoning rule $z = \mu \cdot \rho/(\kappa\, \mu \cdot \beta)$ at the new city's composition minus at the county's before incorporation. Current E grid parameters (provisional). Stated motives: column 2 coding, 60 restrict, 246 none, 16 promote among the 344 model incorporations. SE clustered by county.

Homeowners: if owners do not bear the price of zoning, gains double

Owners gain about twice the estimated gain: in restriction cities 0.22% becomes 0.47%; in promotion cities the loss grows from 0.08% to 0.34%.

Back-of-envelope on the E grid G3-C1 benchmark (additive rung): the housing-cost term of the gain is removed for owners, at each new city's owner share before incorporation (mean 81%) or for everyone. Equal weight per new city. Residents stay put; no landlord income; the remainder is not recomputed.

Back

Homeowners: if owners do not bear the price of zoning, gains double

Gain, % of income	Cities	As estimated	Owners at their share	All owners
All new cities	344	0.12	0.24	0.26
Stated restriction	60	0.22	0.47	0.56
No land use motive	246	0.11	0.21	0.21
Stated promotion	16	−0.08	−0.34	−0.47

Owners gain about twice the estimated gain: in restriction cities 0.22% becomes 0.47%; in promotion cities the loss grows from 0.08% to 0.34%.

Back-of-envelope on the E grid G3-C1 benchmark (additive rung): the housing-cost term of the gain is removed for owners, at each new city's owner share before incorporation (mean 81%) or for everyone. Equal weight per new city. Residents stay put; no landlord income; the remainder is not recomputed.

Back

Homeowners: if owners do not bear the price of zoning, gains double

Gain, % of income	Cities	As estimated	Owners at their share	All owners
All new cities	344	0.12	0.24	0.26
Stated restriction	60	0.22	0.47	0.56
No land use motive	246	0.11	0.21	0.21
Stated promotion	16	−0.08	−0.34	−0.47

In the model every resident pays the higher housing cost that stricter zoning brings, −0.13% of income on average. An owner who stays pays it to herself, so for owners only the zoning amenity remains (+0.17).

Owners gain about twice the estimated gain: in restriction cities 0.22% becomes 0.47%; in promotion cities the loss grows from 0.08% to 0.34%.

Back-of-envelope on the E grid G3-C1 benchmark (additive rung): the housing-cost term of the gain is removed for owners, at each new city's owner share before incorporation (mean 81%) or for everyone. Equal weight per new city. Residents stay put; no landlord income; the remainder is not recomputed.

Back

Fact 2: recent construction predicts incorporation, own growth does not

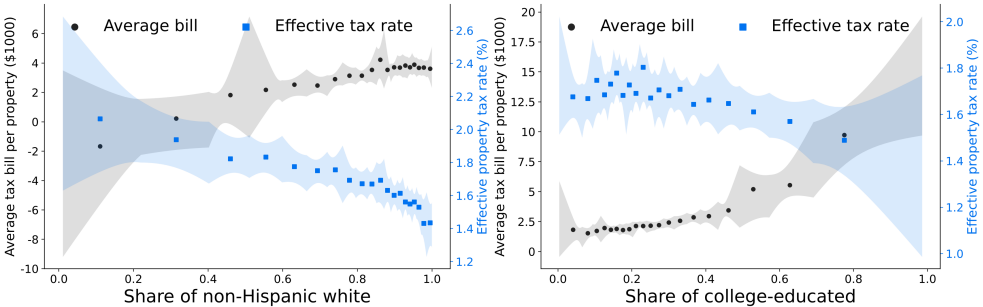
Predictor	Design	Effect (pp)	SE	N / events
Share of homes built in the prior decade (+10 pp)	cross-section	**+0.97**	0.40	1,942 / 314
	panel, per decade	**+0.46**	0.18	6,513 / 318
Housing density (+100 units per km^2)	panel	−0.06	0.04	21,741 / 344
Own housing growth, prior decade (+0.1 log point)	panel	0.00	0.02	15,518 / 196
County growth (+0.1 log point)	panel	0.00	0.02	16,964 / 237
Own minus county growth (+0.1 log point)	panel	+0.01	0.03	15,517 / 196

Places that built more in the decade before are more likely to incorporate; their own past growth is not.

Change in the probability of incorporating (by the end of the period in the cross-section, by the next census in the panel), with the full Fact 2 controls and own home value; SE clustered by county. Assessor records cover new cities and the unincorporated parts of selected counties, so incorporation is more common in these samples (16.2% by the end of the period, 4.9% per decade). Census growth needs a prior count: 148 incorporations drop out of the own-growth rows, 107 of the county-growth row. N: places (cross-section) or place-decades (panel). Bold: significant at 5%.

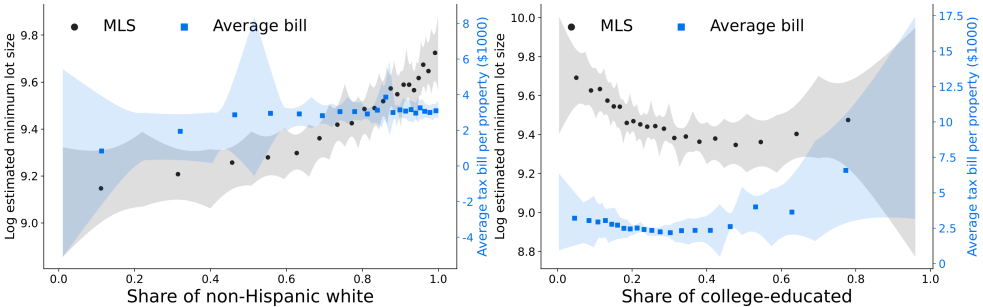
Back

Higher tax bill does not mean higher fiscal burden



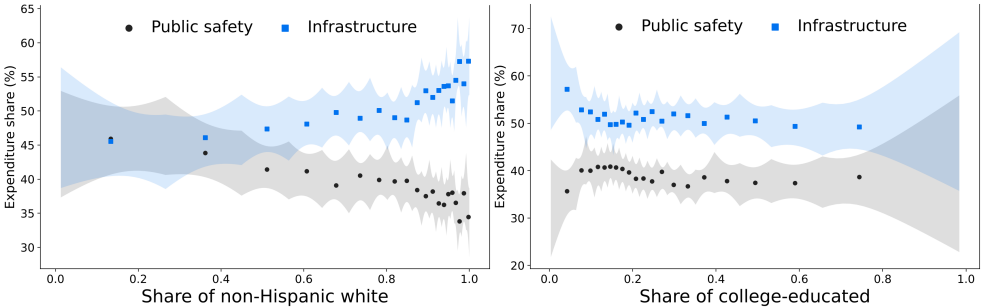
Fact 1

Zoning stringency and property tax bills: home value control



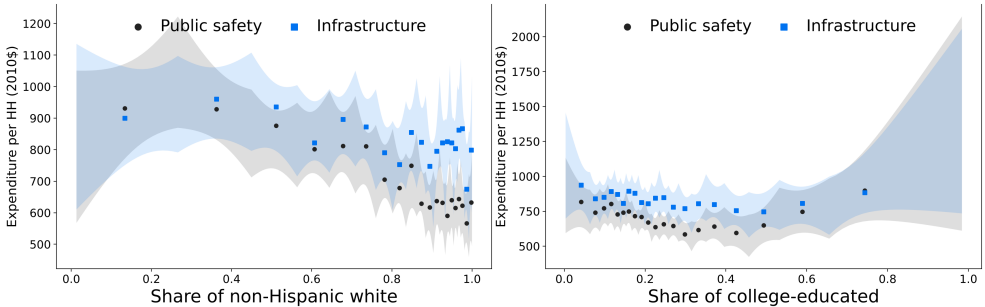
Fact 1

Municipal expenditure share: with home value and demand controls



Fact 1

Municipal exp per HH: with home value and demand controls



Fact 1

Who states which motive?

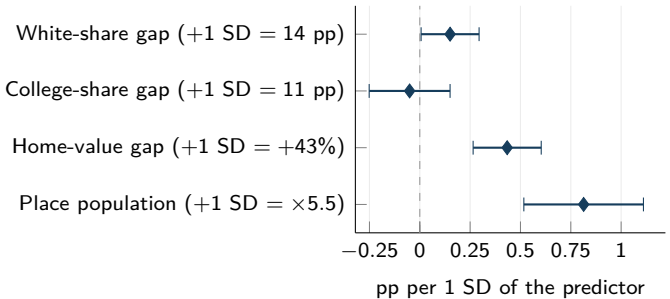
Motives in 353 incorporators and 467 failed attempts

	Public goods	Redistribution	Land use		Others	No record
			Restrictive	Others		
Incorporators	54 (15%)	61 (17%)	63 (18%)	39 (11%)	137 (39%)	178 (50%)
Failed attempts	67 (14%)	67 (14%)	44 (9%)	47 (10%)	140 (30%)	285 (61%)

Percentage points	Public goods	Redistribution	Restrictive land use
College share (+10 pp)	−3.5 (2.5)	−2.9 (2.4)	+0.2 (3.0)
NH White share (+10 pp)	−1.4 (1.0)	+0.9 (0.8)	−2.0 (1.1)
Home value (+10%)	+0.9 (0.6)	**+1.4 (0.7)**	**+3.1 (0.7)**
Population (doubling)	+1.6 (0.9)	**+2.8 (0.9)**	**+4.3 (1.1)**
Housing density (doubling)	**+2.5 (1.0)**	+1.8 (1.2)	−2.4 (1.3)
Mean (%)	15.3	17.3	38.5

Back

Predictors of incorporation: home value gap

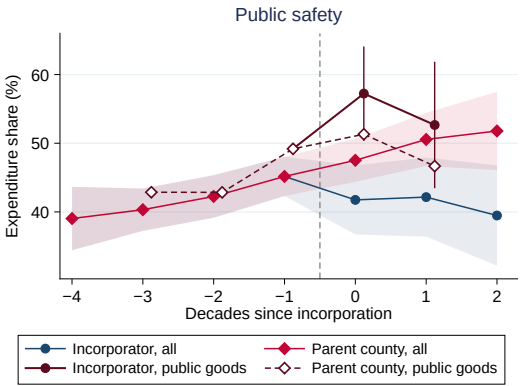


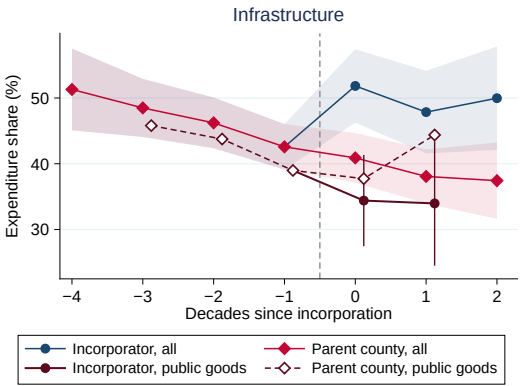
Linear probability model: predictors of incorporation

"at-risk" place-year panel;
gap = lagged gap between place and its unincorporated county

Back

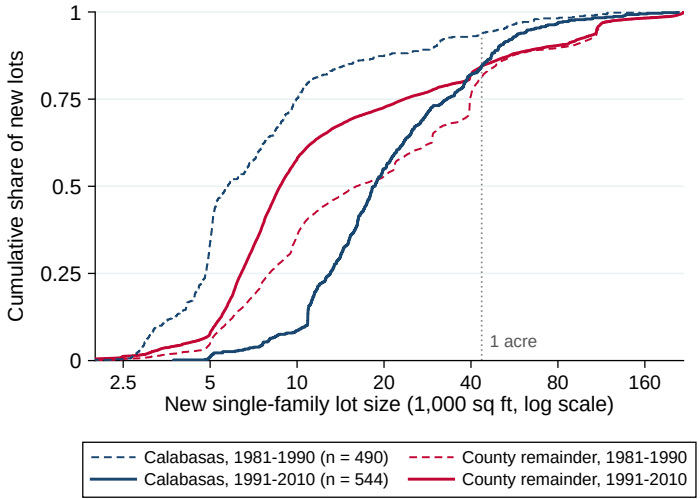
Spending mixes: incorporators with a public goods motive





Back

lot size shifts after incorporation



Calabasas, CA (1991)

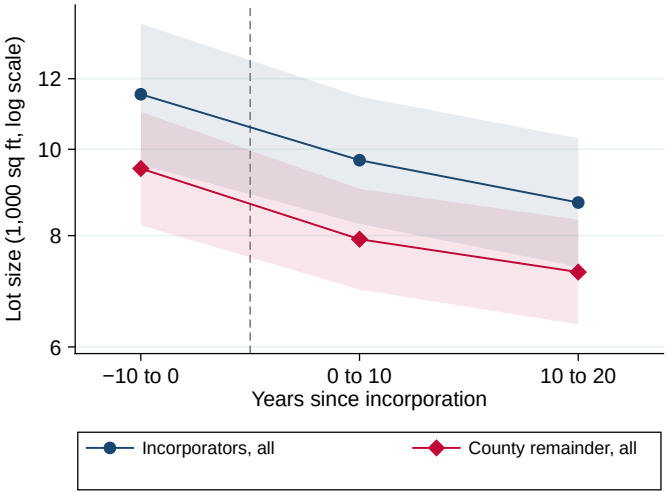
Formed to take local control of growth; 1992 zoning rules against county-approved projects

25th percentile: 4,800 to 13,000 sq ft; county remainder: 8,000 to 6,400

Calabasas General Plan 2030, ch. I; city staff report, Aug. 2018

Back

Option: 10th percentile of new lots



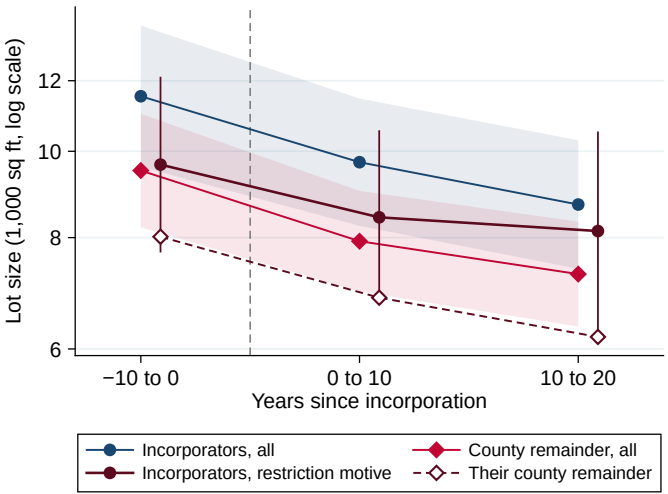
All incorporators (195)

Parallel with the county remainder

Gap change, years 0 to 20: −0.00 log points (SE 0.05); joint p = 0.79

Back

Option: 10th percentile of new lots



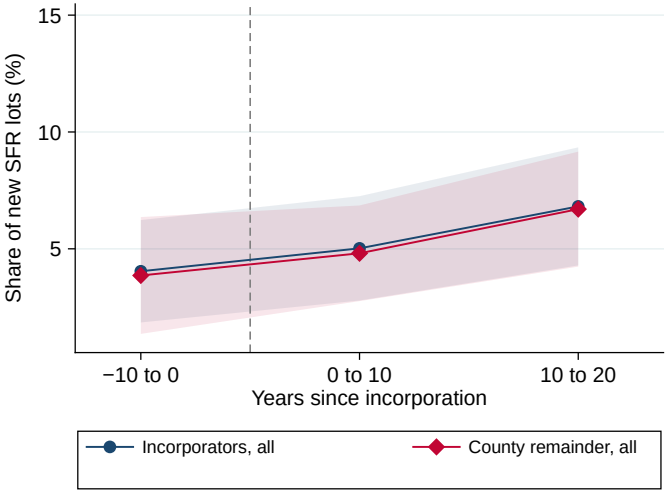
Restriction motive (95)

vs. other incorporators: +0.11 log points (SE 0.09)

Joint test: p = 0.057

Back

Option: new lots under 5,000 sq ft



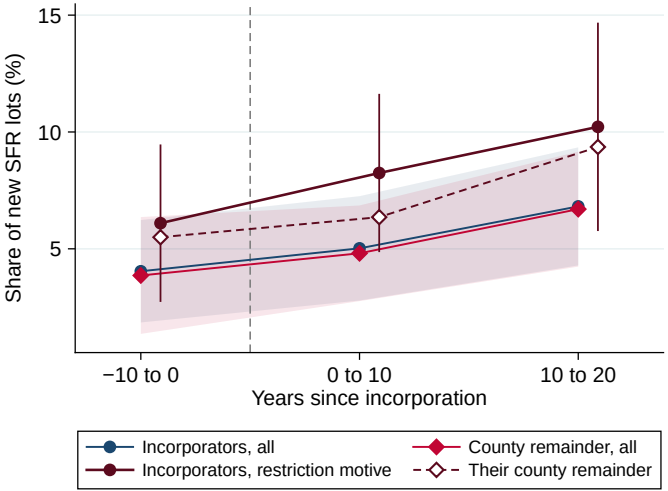
All incorporators (271)

Few new lots are this small: 4 to 10%

Gap change, years 0 to 20: −0.0 pp (SE 0.5); joint p = 0.99

Back

Option: new lots under 5,000 sq ft



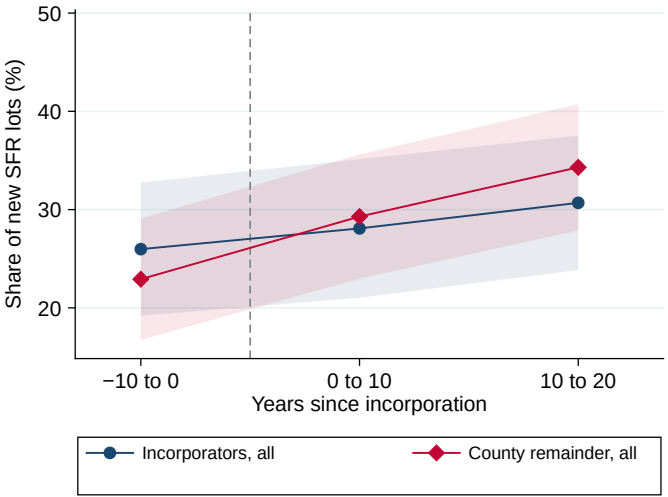
Restriction motive (115)

vs. other incorporators: +1.4 pp (SE 1.2), the wrong sign

Joint test: p = 0.068

Back

Option: new lots under 10,000 sq ft



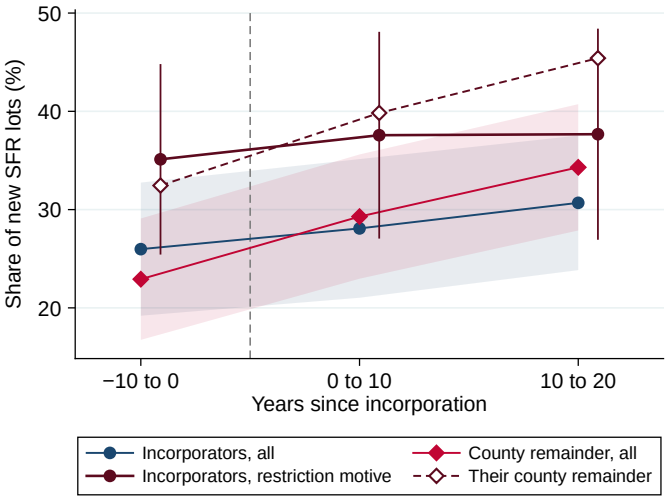
All incorporators (271)

Shrink less than the county remainder: −5.5 pp (SE 1.5), p = 0.002

Places with at least 20 new lots per decade (195): −2.9 pp (SE 1.9), p = 0.26

Back

Option: new lots under 10,000 sq ft



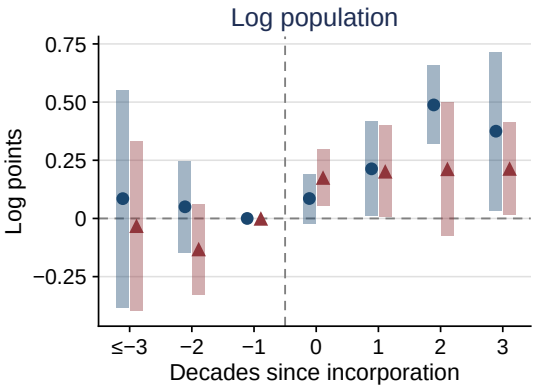
Restriction motive (115)

−7.7 pp (SE 2.2) vs. their county remainder

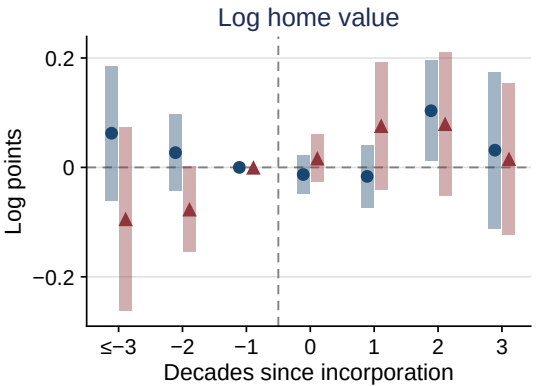
At least 20 lots per decade: −6.7 pp (SE 2.5), p = 0.026

Back

Fact 4 by stated motive: restriction



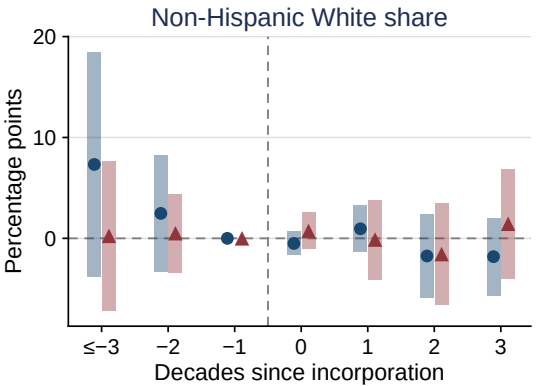
Average: restriction +0.19, others +0.22
Difference −0.03 log points (SE 0.10)



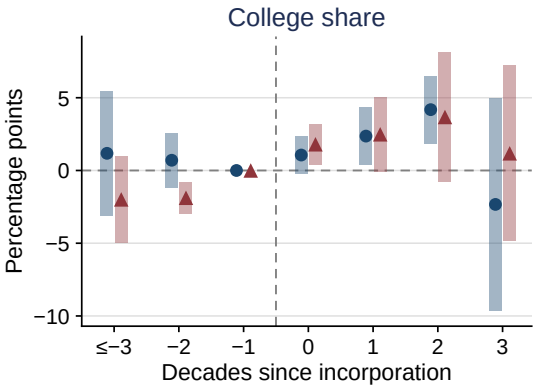
Average: restriction +0.05, others +0.01
Difference +0.04 log points (SE 0.03)

Public goods motive | Table | Back

Fact 4 by stated motive: restriction



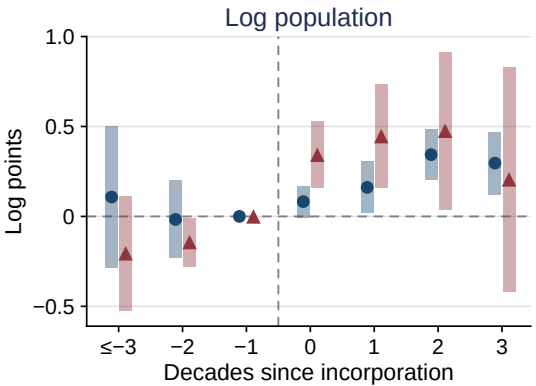
Average: restriction +0.1, others −0.4
Difference +0.5 pp (SE 1.2)



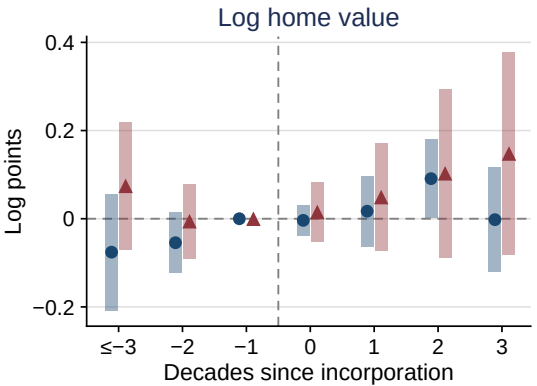
Average: restriction +2.3, others +1.9
Difference +0.3 pp (SE 1.0)

Public goods motive | Table | Back

Fact 4 by stated motive: public goods



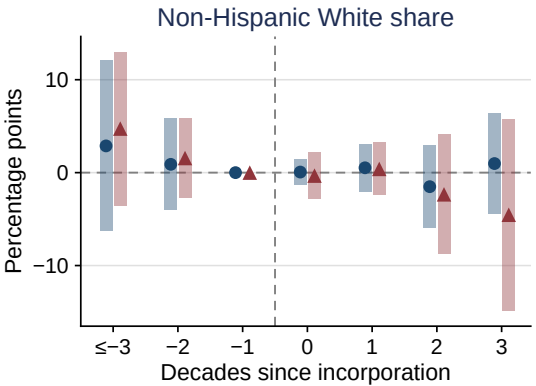
Average: public goods +0.40, others +0.17
Difference +0.23 log points (SE 0.12)



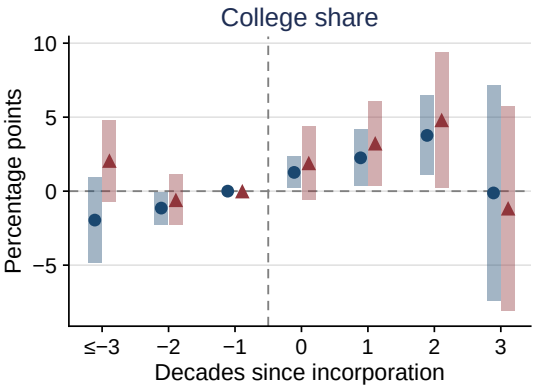
Average: public goods +0.05, others +0.02
Difference +0.03 log points (SE 0.04)

Restriction motive Table Back

Fact 4 by stated motive: public goods



Average: public goods −0.8, others 0.0
Difference −0.7 pp (SE 1.6)



Average: public goods +2.8, others +1.9
Difference +0.9 pp (SE 1.4)

Restriction motive Table Back

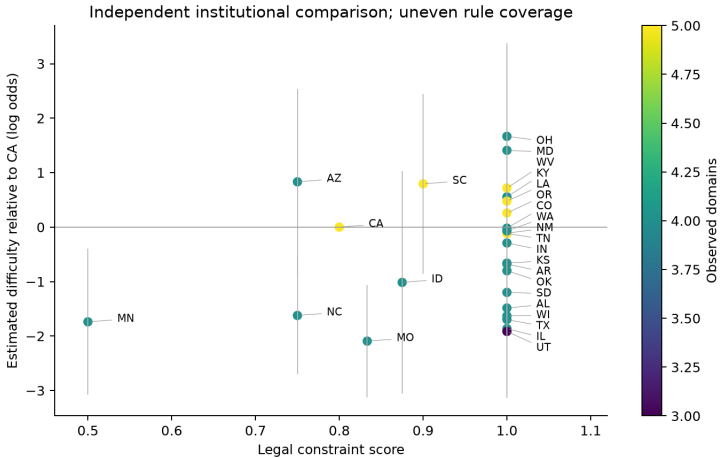


Figure 4: Variations in state-level institutional framework for incorporation

Counterfactual: CA